Southern Nuclear

Regulatory Affairs

3535 Colonnade Parkway
Birmingham, AL 35243
Tel. 205.992.5000

February 12, 2026

Docket Nos.: 52-025
52-026

NL-26-0047
10 CFR 50.90
10 CFR 50.69

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Southern Nuclear Operating Company
Vogtle Electric Generating Plant – Units 3 and 4
Application to Adopt 10 CFR 50.69, "Risk-informed categorization and treatment of structures, systems and components for nuclear power reactors"

Ladies and Gentlemen:

In accordance with the provisions of Sections 50.90 and 50.69 of Title 10 of the *Code of Federal Regulations* (10 CFR), Southern Nuclear Operating Company (SNC) hereby requests an amendment to the combined licenses (COLs) for Vogtle Electric Generating Plant, Unit 3 (License No. NPF-91) and Unit 4 (License No. NPF-92).

The proposed amendment would modify the Vogtle Electric Generating Plant, Units 3 and 4 (VEGP 3&4) licensing basis, by the addition of a license condition, to allow for the implementation of the provisions of 10 CFR 50.69, "Risk-informed categorization and treatment of structures, systems and components for nuclear power reactors." The provisions of 10 CFR 50.69 allow adjustment of the scope of equipment subject to special treatment controls (e.g., quality assurance, testing, inspection, condition monitoring, assessment, and evaluation). For equipment determined to be of low safety significance, alternative treatment requirements can be implemented in accordance with this regulation. For equipment determined to be of high safety significance, requirements will not be changed or will be enhanced. This allows an improved focus on equipment that has greater safety significance resulting in improved plant safety.

The enclosure to this letter provides the basis for the proposed change to the VEGP 3&4 COLs. The categorization process allowed by this change will be consistent with Nuclear Energy Institute (NEI) 00-04, "10 CFR 50.69 SSC Categorization Guideline," Revision 0, dated July 2005, which was endorsed by the Nuclear Regulatory Commission (NRC) in Regulatory Guide 1.201, "Guidelines for Categorizing Structures, Systems, and Components in Nuclear Power Plants According to their Safety Significance," Revision 1, dated May 2006. Attachment 1 of the enclosure provides a list of categorization prerequisites. Use of the categorization process on a plant system will only occur after these prerequisites are met.

The probabilistic risk assessment (PRA) models described within this license amendment request (LAR) are the same as those described within the SNC submittal of the VEGP 3&4 LAR for the adoption of Technical Specifications Task Force (TSTF)-505, Revision 2 (ADAMS Accession No. ML26029A439). SNC requests that the NRC conduct their review of the PRA technical adequacy details for this application in coordination with the review of the TSTF-505 application. This would reduce the SNC and NRC resources necessary to complete the review of the applications. This request should not be considered a linked requested licensing action, as the details of the PRA models in each LAR are complete.

These changes were previously discussed with the NRC staff during a public meeting held on September 17, 2025 (ADAMS Accession No. ML25265A043). SNC has incorporated information into this request to address the topics covered during that meeting. As noted during the discussion, the regulation permits an "applicant" for a combined license (COL) under 10 CFR Part 52 to voluntarily implement the provisions of 10 CFR 50.69; however, the rule does not provide a similar allowance for a "holder" of a Part 52 COL. On February 10, 2026, the NRC granted an exemption from this aspect of the rule, thereby allowing VEGP 3&4, as a Part 52 COL holder, to apply for voluntary application of 10 CFR 50.69 (ADAMS Accession No. ML25262A038).

SNC requests approval of the proposed LAR no later than nine months after acceptance. SNC expects to implement the proposed changes within 90 days of issuance of the amendment.

SNC has concluded that the proposed change presents no significant hazards consideration under the standards set forth in 10 CFR 50.92, "Issuance of amendment."

In accordance with 10 CFR 50.91, SNC is notifying the State of Georgia of this LAR by transmitting a copy of this letter, including the enclosure and attachments, to the designated State official.

This letter contains no regulatory commitments. This letter has been reviewed and determined not to contain security-related information.

If you have any questions, please contact Ryan Joyce at 205.992.6468.

I declare under penalty of perjury that the foregoing is true and correct.
Executed on the 12th day of February 2026.

Respectfully submitted,

Jamie Coleman

Digitally signed by JAMIEMCO
Date: 2026.02.12 15:34:35 -06'00'

Jamie M. Coleman
Regulatory Affairs Director

Enclosure: Evaluation of the Proposed Change

cc: NRC Regional Administrator, Region II
NRC Project Manager – Vogtle 3&4
Senior Resident Inspector – Vogtle 3&4
Director, Environmental Protection Division – State of Georgia
Document Services RTYPE: VND.LI.L00

Enclosure

Evaluation of the Proposed Change

Subject: Application to Adopt 10 CFR 50.69, "Risk-informed categorization and treatment of structures, systems and components for nuclear power reactors"

1. SUMMARY DESCRIPTION
2. DETAILED DESCRIPTION
   - 2.1 Current Regulatory Requirements
   - 2.2 Reason for the Proposed Change
   - 2.3 Description of the Proposed Change
3. TECHNICAL EVALUATION
   - 3.1 Categorization Process Description (10 CFR 50.69(b)(2)(i))
     - 3.1.1 Overall Categorization Process
     - 3.1.2 Passive Categorization Process
   - 3.2 Technical Adequacy Evaluation (10 CFR 50.69(b)(2)(ii))
     - 3.2.1 Internal Events and Internal Flooding
     - 3.2.2 Fire Hazards
     - 3.2.3 Seismic Hazards
     - 3.2.4 Other External Hazards
     - 3.2.5 Low Power and Shutdown
     - 3.2.6 PRA Maintenance and Updates
     - 3.2.7 PRA Uncertainty Evaluations
   - 3.3 PRA Review Process Results (10 CFR 50.69(b)(2)(iii))
   - 3.4 Risk Evaluations (10 CFR 50.69(b)(2)(iv))
   - 3.5 Feedback and Adjustment Process
4. REGULATORY EVALUATION
   - 4.1 Applicable Regulatory Requirements/Criteria
   - 4.2 No Significant Hazards Consideration Analysis
   - 4.3 Conclusions
5. ENVIRONMENTAL CONSIDERATION
6. REFERENCES

ATTACHMENTS:

1. List of Categorization Prerequisites
2. Description of PRA Models Used in Categorization
3. Disposition and Resolution of Open Peer Review Findings and Self-Assessment Open Items
4. External Hazards Screening
5. Progressive Screening Approach for Addressing External Hazards
6. Disposition of Key Assumptions/Sources of Uncertainty

1. SUMMARY DESCRIPTION

The proposed amendment modifies the licensing basis to allow for the implementation of the provisions of Title 10 of the *Code of Federal Regulations* (10 CFR), Section 50.69, "Risk-informed categorization and treatment of structures, systems and components for nuclear power reactors." The provisions of 10 CFR 50.69 allow adjustment of the scope of equipment subject to special treatment controls (e.g., quality assurance, testing, inspection, condition monitoring, assessment, and evaluation). For equipment determined to be of low safety significance, alternative treatment requirements can be implemented in accordance with this regulation. For equipment determined to be of high safety significance, requirements will not be changed or will be enhanced. This allows an improved focus on equipment that has greater safety significance resulting in improved plant safety.

2. DETAILED DESCRIPTION

2.1 Current Regulatory Requirements

The Nuclear Regulatory Commission (NRC) has established a set of regulatory requirements for commercial nuclear reactors to ensure that a reactor facility does not impose an undue risk to the health and safety of the public, thereby providing reasonable assurance of adequate protection to public health and safety. The current body of NRC regulations and their implementation are largely based on a "deterministic" approach.

This deterministic approach establishes requirements for engineering margin and quality assurance in design, manufacture, and construction. In addition, it assumes that adverse conditions can exist (e.g., equipment failures and human errors) and establishes a specific set of design basis events (DBEs). The deterministic approach then requires that the facility include safety systems capable of preventing or mitigating the consequences of those DBEs to protect public health and safety. The structures, systems and components (SSCs) necessary to defend against the DBEs are defined as "safety-related," and these SSCs are the subject of many regulatory requirements, herein referred to as "special treatments," designed to ensure that they are of high quality and high reliability, and have the capability to perform during postulated design basis conditions. Treatment includes, but is not limited to, quality assurance, testing, inspection, condition monitoring, assessment, evaluation, and resolution of deviations. The distinction between "treatment" and "special treatment" is the degree of NRC specification as to what must be implemented for particular SSCs or conditions. Typically, the regulations establish the scope of SSCs that receive special treatment using one of three different terms: "safety-related," "important to safety," or "basic component." The terms "safety-related" and "basic component" are defined in the regulations, while "important to safety," used principally in the general design criteria (GDC) of Appendix A to 10 CFR Part 50, is not explicitly defined.

Paragraph (b) of 10 CFR 50.69 defines the applicability and scope of the risk-informed treatment of SSCs and establishes the associated submittal and approval process. Under the rule, an "applicant" for a combined license (COL) under 10 CFR Part 52 may voluntarily implement the provisions of 10 CFR 50.69; however, the rule does not provide a similar allowance for a "holder" of a Part 52 COL. On February 10, 2026, the NRC granted an exemption from this aspect of the rule, thereby allowing VEGP 3&4, as a Part 52 COL holder, to apply for voluntary application of 10 CFR 50.69 (Reference 6).

2.2 Reason for the Proposed Change

A probabilistic approach to regulation enhances and extends the traditional deterministic approach by allowing consideration of a broader set of potential challenges to safety, providing a logical means for prioritizing these challenges based on safety significance, and allowing consideration of a broader set of resources to defend against these challenges. In contrast to the deterministic approach, probabilistic risk assessments (PRAs) address credible initiating events by assessing the event frequency. Mitigating system reliability is then assessed, including the potential for common cause failures. The probabilistic approach to regulation is an extension and enhancement of traditional regulation by considering risk in a comprehensive manner.

To take advantage of the safety enhancements available using PRA, in 2004 the NRC published a new regulation, 10 CFR 50.69. The provisions of 10 CFR 50.69 allow adjustment of the scope of equipment subject to special treatment (e.g., quality assurance, testing, inspection, condition monitoring, assessment, and evaluation). For equipment determined to be Low Safety Significant (LSS), alternative treatment requirements can be implemented in accordance with the regulation. For equipment determined to be High Safety Significant (HSS), requirements will not be changed or will be enhanced. This allows an improved focus on equipment that has greater safety significance resulting in improved plant safety.

The rule contains requirements on how a licensee categorizes SSCs using a risk-informed process, adjusts treatment requirements consistent with the relative significance of the SSC, and manages the process over the lifetime of the plant. A risk-informed categorization process is employed to determine the safety significance of SSCs and place the SSCs into one of four risk-informed safety class (RISC) categories. The determination of safety significance is performed by an integrated decision-making process, as described by Nuclear Energy Institute (NEI) 00-04, "10 CFR 50.69 SSC Categorization Guideline" (Reference 1), which uses both risk insights and traditional engineering insights. The safety functions include the design basis functions, as well as functions credited for severe accidents (including external events). Special or alternative treatment for the SSCs is applied as necessary to assure functionality and reliability are maintained and is a function of the SSC categorization results and associated bases. Finally, periodic assessment activities are conducted to adjust the categorization and/or treatment processes as needed so that SSCs continue to meet all applicable requirements.

The rule does not allow for the elimination of SSC functional requirements or allow equipment that is required by the deterministic design basis to be removed from the facility. Instead, the rule enables licensees to focus their resources on SSCs that make a significant contribution to plant safety. For SSCs that are categorized as HSS, existing treatment requirements are maintained or enhanced. Conversely, for SSCs that do not significantly contribute to plant safety on an individual basis, the rule allows an alternative risk-informed approach to treatment that provides reasonable confidence (which is a reduced level compared to the reasonable assurance criteria used for many special treatments) that these SSCs will satisfy functional requirements.

Implementation of 10 CFR 50.69 will allow Southern Nuclear Operating Company (SNC) to improve the focus on equipment that has greater safety significance resulting in improved plant safety.

2.3 Description of the Proposed Change

SNC proposes the addition of the following condition to the combined licenses (COLs) of Vogtle Electric Generating Plant, Units 3 and 4 (VEGP 3&4) to document the NRC's approval of the use of 10 CFR 50.69.

> SNC is approved to implement 10 CFR 50.69 using the processes for categorization of Risk-Informed Safety Class (RISC)-1, RISC-2, RISC-3, and RISC-4 Structures, Systems, and Components (SSCs) using: Probabilistic Risk Assessment (PRA) models to evaluate risk associated with internal events, including internal flooding, and internal fire; the shutdown safety assessment process to assess shutdown risk; the Arkansas Nuclear One, Unit 2 (ANO-2) passive categorization method to assess passive component risk for Class 2 and Class 3 and non-Class SSCs and their associated supports; the results of the non-PRA evaluations that are based on the IPEEE Screening Assessment for External Hazards updated using the external hazard screening significance process identified in ASME/ANS PRA Standard RA-Sa-2009 for other external hazards except seismic; and the alternative seismic approach as described in SNC's submittal letter dated [DATE], and all its subsequent associated supplements as specified in License Amendment No. [XXX] dated [DATE].
>
> Prior NRC approval, under 10 CFR 50.90, is required for a change to the categorization process specified above.

3. TECHNICAL EVALUATION

10 CFR 50.69 specifies the information to be provided by a licensee requesting adoption of the regulation. This request conforms to the requirements of 10 CFR 50.69(b)(2), which states:

> A licensee voluntarily choosing to implement this section shall submit an application for license amendment under § 50.90 that contains the following information:
>
> (i) A description of the process for categorization of RISC–1, RISC–2, RISC–3 and RISC–4 SSCs.
>
> (ii) A description of the measures taken to assure that the quality and level of detail of the systematic processes that evaluate the plant for internal and external events during normal operation, low power, and shutdown (including the plant-specific probabilistic risk assessment (PRA), margins-type approaches, or other systematic evaluation techniques used to evaluate severe accident vulnerabilities) are adequate for the categorization of SSCs.
>
> (iii) Results of the PRA review process conducted to meet § 50.69(c)(1)(i).
>
> (iv) A description of, and basis for acceptability of, the evaluations to be conducted to satisfy § 50.69(c)(1)(iv). The evaluations must include the effects of common cause interaction susceptibility, and the potential impacts from known degradation mechanisms for both active and passive functions, and address internally and externally initiated events and plant operating modes (e.g., full power and shutdown conditions).

Each of these submittal requirements are addressed in the following sections.

The PRA models described within this license amendment request (LAR) are the same as those described within the SNC submittal of the VEGP 3&4 LAR for the adoption of Technical Specifications Task Force (TSTF)-505, Revision 2 (Reference 5). SNC requests that the NRC conduct their review of the PRA technical adequacy details for this application in coordination with the review of the TSTF-505 application. This would reduce the SNC and NRC resources necessary to complete the review of the applications. This request should not be considered a linked requested licensing action, as the details of the PRA models in each LAR are complete.

3.1 Categorization Process Description (10 CFR 50.69(b)(2)(i))

3.1.1 Overall Categorization Process

SNC will implement the risk categorization process in accordance with NEI 00-04, Revision 0, as endorsed by Regulatory Guide (RG) 1.201, "Guidelines for Categorizing Structures, Systems, and Components in Nuclear Power Plants According to their Safety Significance" (Reference 2). NEI 00-04 Section 1.5 states "Due to the varying levels of uncertainty and degrees of conservatism in the spectrum of risk contributors, the risk significance of SSCs is assessed separately from each of five risk perspectives and used to identify SSCs that are potentially safety-significant." A separate evaluation is appropriate to avoid reliance on a combined result that may mask the results of individual risk contributors.

The process to categorize each system will be consistent with the guidance in NEI 00-04, as endorsed by RG 1.201, with the exception of the seismic hazard impact evaluation, which will use the Electric Power Research Institute (EPRI) 3002017583 (Reference 3) approach for seismic Tier 1 sites. The inclusion of additional process steps to address seismic considerations will ensure that the reasonable confidence required by 10 CFR 50.69(c)(1)(iv) is achieved. RG 1.201 states that "the implementation of all processes described in NEI 00-04 (i.e., Sections 2 through 12) is integral to providing reasonable confidence" and that "all aspects of NEI 00-04 must be followed to achieve reasonable confidence in the evaluations required by § 50.69(c)(1)(iv)." However, neither RG 1.201 nor NEI 00-04 prescribe a particular sequence for completing the categorization elements. Therefore, the order in which each element of the categorization process (listed below) is completed is flexible as long as all five elements are completed; they may even be performed in parallel. Note that NEI 00-04 only requires Item 3 (seven qualitative criteria) to be completed for components or functions categorized as LSS by all other elements. Similarly, Item 4 (Defense-in-Depth assessment) is only required for safety-related active components or functions categorized as LSS by all other elements.

1. PRA-based evaluations (e.g., the internal events, internal flooding, and fire PRAs)
2. Non-PRA approaches (e.g., other external events screening, and shutdown assessment)
3. Seven qualitative criteria in Section 9.2 of NEI 00-04
4. Defense-in-Depth (DID) assessment
5. Passive categorization methodology

Figure 3-1 illustrates the major steps of the categorization process described in NEI 00-04. As explained further in Section 3.2.3, two categorization steps (represented by four blocks on the figure) have been included to highlight review of seismic insights as they pertain to this application.

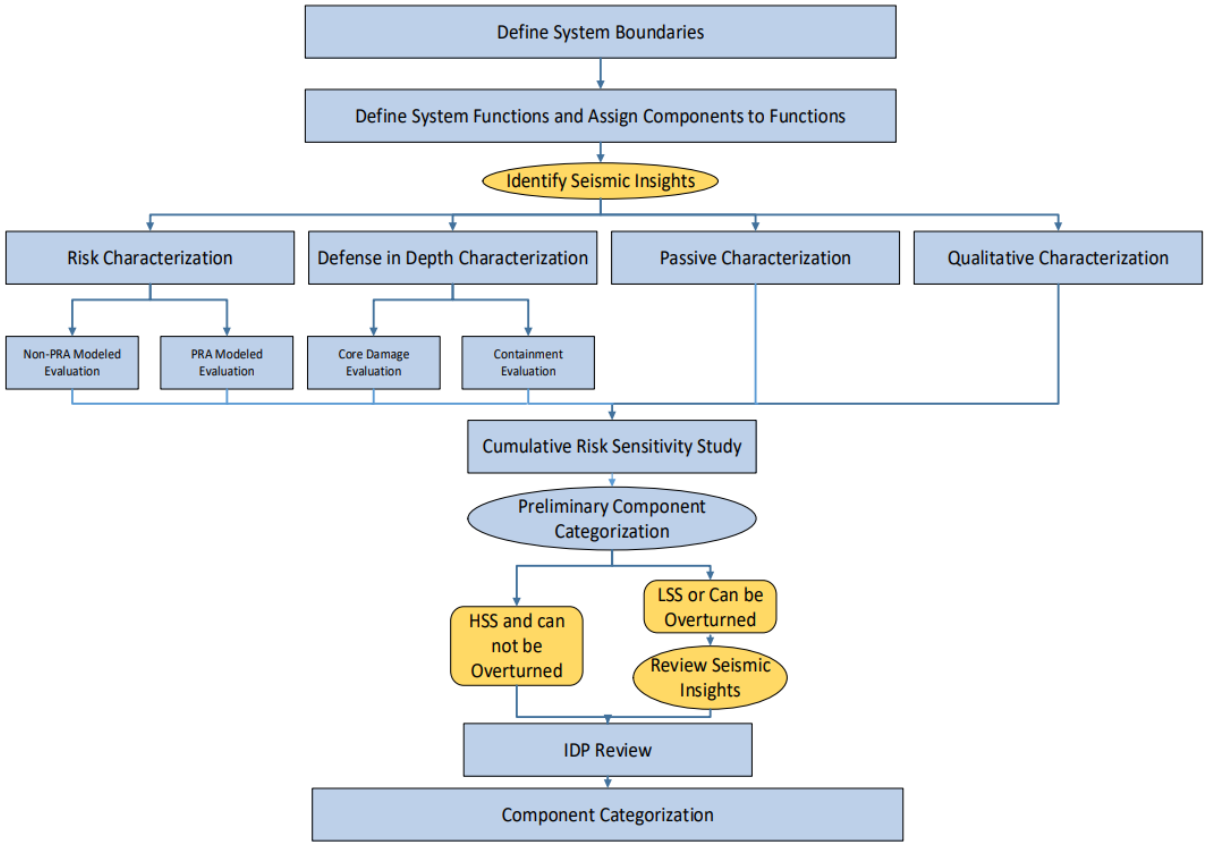

Figure 3-1: Categorization Process Overview

Categorization of SSCs will be completed per the NEI 00-04 process, as endorsed by RG 1.201, which includes the determination of safety significance through the various elements identified above. The results of these elements are used as inputs to arrive at a preliminary component categorization (i.e., HSS or LSS) that is presented to the Integrated Decision-Making Panel (IDP). Note that the term "preliminary HSS or LSS" used in this application is synonymous with the NEI 00-04 term "candidate HSS or LSS." A component or function is preliminarily categorized as HSS if any element of the process results in a preliminary HSS determination in accordance with Table 3-1. The safety significance determination of each element is independent and therefore the sequence of the elements does not impact the resulting preliminary categorization of each component or function. Consistent with NEI 00-04, the categorization of a component or function will only be "preliminary" until it has been confirmed by the IDP. Once the IDP confirms that the categorization process was followed appropriately, the final RISC category can be assigned.

The IDP may direct and approve detailed categorization of components in accordance with NEI 00-04, Section 10.2. The IDP may always elect to change a preliminary LSS component or function to HSS, however the ability to change component categorization from preliminary HSS to LSS is limited. This ability is only available to the IDP for select process steps as described in NEI 00-04 and endorsed by RG 1.201. Table 3-1 summarizes the NEI 00-04 IDP limitations. The steps of the process are performed at either the function level, component level, or both. This is also summarized in Table 3-1. A component is assigned its final RISC category upon approval by the IDP.

Table 3-1: Categorization Evaluation Summary

Element	Categorization Step – NEI 00-04 Section	Evaluation Level	IDP Change HSS to LSS	Drives Associated Functions
Risk (PRA Modeled)	Internal Events Base Case – Section 5.1	Component	Not Allowed	Yes
	Fire, Seismic and Other External Events Base Case		Allowable	No
	PRA Sensitivity Studies		Allowable	No
	Integral PRA Assessment – Section 5.6		Not Allowed	Yes
Risk (Non-modeled)	Fire and Other External Hazards	Component	Not Allowed	No
	Seismic – Alternative Tier 1 Approach	Function/Component	Allowed [1]	No
	Shutdown – Section 5.5	Function/Component	Not Allowed	No
Defense-in-Depth	Core Damage – Section 6.1	Function/Component	Not Allowed	Yes
	Containment – Section 6.2	Component	Not Allowed	Yes
Qualitative Criteria	Considerations – Section 9.2	Function	Allowable [2]	N/A
Passive	Passive – Section 4	Segment/Component	Not Allowed	No

Notes:
[1] IDP consideration of seismic insights can also result in an LSS to HSS determination.

[2] The assessments of the qualitative considerations are agreed upon by the IDP in accordance with NEI 00-04, Section 9.2. In some cases, a 10 CFR 50.69 categorization team may provide preliminary assessments of the seven considerations for the IDP's consideration, however the final assessments of the seven considerations are the direct responsibility of the IDP.

(Note 2 continued)
The seven considerations are addressed preliminarily by the 10 CFR 50.69 categorization team for at least the system functions that are not found to be HSS due to any other categorization step. Each of the seven considerations requires a supporting justification for confirming (true response) or not confirming (false response) that consideration. If the 10 CFR 50.69 categorization team determines that one or more of the seven considerations cannot be confirmed, then that function is presented to the IDP as preliminary HSS. Conversely, if all the seven considerations are confirmed, then the function is presented to the IDP as preliminary LSS.

The System Categorization Document, including the justifications provided for the qualitative considerations, is reviewed by the IDP. The IDP is responsible for reviewing the preliminary assessment to the same level of detail as the 10 CFR 50.69 team (i.e., all considerations for all functions are reviewed). The IDP may confirm the preliminary function risk and associated justification or may direct that it be changed based upon their expert knowledge. Because the Qualitative Criteria are the direct responsibility of the IDP, changes may be made from preliminary HSS to LSS or from preliminary LSS to HSS at the discretion of the IDP. If the IDP determines any of the seven considerations cannot be confirmed (false response) for a function, then the final categorization of that function is HSS.

The mapping of components to system functions is used in some categorization process steps to facilitate preliminary categorization of components. Specifically, functions with mapped components that are determined to be HSS by the PRA-based assessment (i.e., internal events PRA or integral PRA assessment) or DID evaluation will be initially treated as HSS. However, NEI 00-04, Section 10.2, allows detailed categorization which can result in some components mapped to HSS functions being treated as LSS. Also, Section 4.0 discusses additional functions that may be identified (e.g., fill and drain) to group and consider potentially LSS components that may have been initially associated with an HSS function, but do not support the critical attributes of that HSS function. Note that certain steps of the categorization process are performed at a component level (e.g., Passive, Non-PRA-modeled hazards – see Table 3-1). Except for seismic, these components from the component level assessments will remain HSS (IDP cannot override) regardless of the significance of the functions to which they are mapped. Components having seismic functions may be HSS or LSS based on the IDP's consideration of the seismic insights applicable to the system being categorized. Therefore, if an HSS component is mapped to an LSS function, that component will remain HSS. If an LSS component is mapped to an HSS function, that component may be driven HSS based on Table 3-1 or may remain LSS. Given that VEGP 3&4 is a seismic Tier 1 (low seismic hazard) plant as defined in Reference 3, seismic considerations are not required to drive an HSS determination at the component level, but the IDP will consider available seismic information pertinent to the components being categorized and can, at its discretion, determine that a component should be HSS based on that information.

The following clarifications are applied to the NEI 00-04 categorization process:

- The IDP will be composed of a group of at least five experts who collectively have expertise in plant operations, design engineering, systems engineering, safety analysis, and PRA. The makeup of the IDP will follow Section 9.1 of NEI 00-04.
- The IDP will be trained in the specific technical aspects and requirements related to the categorization process. Training will address, at a minimum, (1) the purpose of the categorization; (2) present treatment requirements for SSCs including requirements for

DBEs; (3) PRA fundamentals; (4) details of the plant-specific PRA including the modeling, scope, and assumptions, the interpretation of risk importance measures, and the role of sensitivity studies and the change-in-risk evaluations; and (5) the DID philosophy and requirements to maintain this philosophy.

- The decision criteria for the IDP to categorize SSCs as HSS or LSS in accordance with § 50.69(f)(1) will be documented in SNC procedures.
- Decisions of the IDP will be arrived at by consensus. Differing opinions will be documented and resolved, if possible. However, a simple majority of the panel is sufficient for final decisions regarding HSS and LSS.
- Passive characterization will be performed using the processes described in Section 3.1.2 of this enclosure. Consistent with NEI 00-04, an HSS determination by the passive categorization process cannot be changed by the IDP.
- An unreliability factor of 3 will be used for the sensitivity studies described in Section 8 of NEI 00-04. The factor of 3 was chosen as it is representative of the typical error factor of basic events used in the PRA model.
- NEI 00-04, Section 7, requires assigning the safety significance of functions to be preliminary HSS if it is supported by an SSC determined to be HSS from the PRA-based assessment in Section 5, but does not require this for SSCs determined to be HSS from non-PRA-based, deterministic assessments in Section 5. This requirement is further clarified in the VEGP 1&2 Safety Evaluation (Reference 4) which states "…if any SSC is identified as HSS from either the integrated PRA component safety significance assessment (Section 5 of NEI 00-04) or the [DID] assessment (Section 6 of NEI 00-04), the associated system function(s) would be identified as HSS."
- Once a system function is identified as HSS, then all the components that support that function are preliminary HSS. The IDP must intervene to assign any of these HSS function components to LSS.
- With regard to the criteria that consider whether the active function is called out or relied upon in the plant Emergency/Abnormal Operating Procedures, SNC will not take credit for alternate means unless the alternate means are proceduralized and included in Licensed Operator training.
- SNC proposes to apply an alternative seismic approach to those listed in NEI 00-04 Sections 1.5 and 5.3, as discussed in Section 3.2.3 of this enclosure.

The risk analysis to be implemented for each modeled hazard is described below.

- Internal Event Risks: Internal events and internal flooding PRA, as submitted to the NRC for TSTF-505 (Reference 5).
- Fire Risks: Fire PRA model, as submitted to the NRC for TSTF-505 (Reference 5).
- Seismic Risks: Alternative approach in EPRI 3002017583 (Reference 3) for Tier 1 plants and additional considerations discussed in Section 3.2.3 of this enclosure.
- Other External Risks (e.g., tornados, external floods): Using the Individual Plant Examination of External Events (IPEEE) screening process. The other external hazards were determined to be insignificant contributors to plant risk.

- Low Power and Shutdown Risks: Qualitative DID shutdown model for shutdown configuration risk management (CRM) based on the framework for DID provided in NUMARC 91-06, "Guidance for Industry Actions to Assess Shutdown Management" (Reference 7), which provides guidance for assessing and enhancing safety during shutdown operations.

A change to the categorization process that is outside the bounds specified above (e.g., change from the Tier 1 alternative seismic approach to the Tier 2 approach) will not be made without prior NRC approval. The SSC categorization process documentation will include the following elements:

1. Program procedures used in the categorization
2. System functions, identified and categorized with the associated bases
3. Mapping of components to support function(s)
4. PRA model results, including sensitivity studies
5. Hazards analyses, as applicable
6. Passive categorization results and bases
7. Categorization results including all associated bases and RISC classifications
8. Component critical attributes for HSS SSCs
9. Results of periodic reviews and SSC performance evaluations
10. IDP meeting minutes and qualification/training records for the IDP members

3.1.2 Passive Categorization Process

For the purposes of 10 CFR 50.69 categorization, passive components are those components that have a pressure retaining function. Passive components and the passive function of active components will be evaluated using the Arkansas Nuclear One (ANO) Risk-Informed Repair/Replacement Activities (RI-RRA) methodology contained in Reference 8 consistent with the related Safety Evaluation (SE) issued by the NRC.

The RI-RRA methodology is a risk-informed safety classification and treatment program for repair/replacement activities for pressure retaining items and their associated supports. In this method, the component failure is assumed with a probability of 1.0 and only the consequence evaluation is performed. It additionally applies deterministic considerations (e.g., DID, safety margins) in determining safety significance. Component supports, if categorized, are assigned the same safety significance as the highest passively ranked component within the bounds of the associated analytical pipe stress model. Consistent with NEI 00-04, an HSS determination by the passive categorization process cannot be changed by the IDP.

The use of this method for a 10 CFR 50.69 application was previously approved by the NRC as documented in the final SE for VEGP 1&2, dated December 17, 2014 (Reference 4). The RI-RRA method as approved for use at VEGP 1&2 for 10 CFR 50.69 does not have any plant specific aspects and is generic. It relies on the conditional core damage and large early release probabilities associated with postulated ruptures. Safety significance is generally measured by the frequency and the consequence of the event. However, this RI-RRA process categorizes components solely based on consequence, which measures the safety significance of the passive component given that it ruptures. This approach is conservative compared to including the rupture frequency in the categorization since this approach will not allow the categorization of SSCs to be affected by any changes in frequency due to changes in treatment. The passive

categorization process is intended to apply the same risk-informed process accepted by the NRC in Reference 8 for the passive categorization of Class 2, 3, and non-class components. This is the same passive SSC scope the NRC has conditionally endorsed in ASME Code Cases N-660 and N-662 as published in RG 1.147, "Inservice Inspection Code Case Acceptability, ASME Section XI, Division 1," Revision 15 (Reference 9). Both code cases employ a similar risk-informed safety classification of SSCs in order to change the repair/replacement requirements of the affected LSS components. All ASME Code Class 1 SSCs with a pressure retaining function, as well as supports, will be assigned HSS for passive categorization which will result in HSS for its risk-informed safety classification and cannot be changed by the IDP. Therefore, this methodology and scope for passive categorization is acceptable and appropriate for use at VEGP 3&4 for 10 CFR 50.69 SSC categorization.

3.2 Technical Adequacy Evaluation (10 CFR 50.69(b)(2)(ii))

The following sections demonstrate that the quality and level of detail of the processes used in categorization of SSCs are adequate. The PRA models described below have been peer reviewed and there are no PRA upgrades that have not been peer reviewed. The PRA models credited in this request are the same PRA models credited in the VEGP 3&4 application to adopt TSTF-505 (Reference 5).

3.2.1 Internal Events and Internal Flooding

The VEGP 3&4 categorization process for the internal events and flooding hazard will use the plant-specific PRA model. The SNC risk management process ensures that the PRA model used in this application reflects the as-built and as-operated plant for VEGP 3&4. Attachment 2 of this enclosure identifies the applicable internal events and internal flooding PRA models.

3.2.2 Fire Hazards

The VEGP 3&4 categorization process for fire hazards will use a peer reviewed plant-specific fire PRA model. The internal Fire PRA model was developed consistent with NUREG/CR-6850 and only utilizes methods previously accepted by the NRC. The SNC risk management process ensures that the PRA model used in this application reflects the as-built and as-operated plant for VEGP 3&4. Attachment 2 of this enclosure identifies the applicable fire PRA model.

3.2.3 Seismic Hazards

10 CFR 50.69(c)(1) requires the use of PRA to assess risk from internal events. For other risk hazards, such as seismic, 10 CFR 50.69(b)(2) allows, and NEI 00-04 summarizes the use of other methods for determining SSC functional importance in the absence of a quantifiable PRA (such as Seismic Margin Analysis or IPEEE Screening) as part of an integrated, systematic process. For the VEGP 3&4 seismic hazard assessment, SNC proposes to use a risk-informed graded approach that meets the requirements of 10 CFR 50.69(b)(2) as an alternative to those listed in NEI 00-04 Sections 1.5 and 5.3. The approach is specified in EPRI 3002017583, "Alternative Approaches for Addressing Seismic Risk in 10 CFR 50.69 Risk-Informed Categorization" (Reference 3) and includes additional qualitative considerations provided in this section.

The VEGP 3&4 site meets the EPRI 3002017583 Tier 1 criteria for a "Low Seismic Hazard/High Seismic Margin" site. The Tier 1 criteria are as follows:

> Tier 1: Plants where the GMRS (Ground Motion Response Spectrum) peak acceleration is at or below approximately 0.2g or where the GMRS is below or approximately equal to the SSE (Safe Shutdown Earthquake) between 1.0 Hz and 10 Hz. Examples are shown in Figures 2-1 and 2-2. At these sites, the GMRS is either very low or within the range of the SSE such that unique seismic categorization insights are not expected.

Note: Reference 3 applies to the Tier 1 sites in its entirety except for Sections 2.3 (Tier 2 sites), 2.4 (Tier 3 sites), Appendix A (seismic correlation), and Appendix B (criteria for capacity-based screening).

The Tier 1 criterion (i.e., basis) in EPRI 3002017583 is a comparison of the ground motion response spectrum (GMRS, derived from the seismic hazard) to the safe shutdown earthquake (SSE, i.e., seismic design basis capability). U.S. nuclear power plants that utilize the 10 CFR 50.69 Seismic Alternative (EPRI 3002017583) will continue to compare GMRS to SSE.

The trial studies in EPRI 3002017583 show that seismic categorization insights are overlaid by other risk insights even at plants where the GMRS is far beyond the seismic design basis. Therefore, the basis for the Tier 1 classification and resulting criteria is not that the design basis insights are adequate. Instead, consideration of the full range of the seismic hazard produces limited unique insights to the categorization process. That is the basis for the following statements in Table 4-1 of the EPRI report.

> At Tier 1 sites, the likelihood of identifying a unique seismic condition that would cause an SSC to be designated HSS is very low. Therefore, with little to no anticipated unique seismic insights, the 50.69 categorization process using the FPIE [full power internal events] PRA and other risk evaluations along with the required [DID] and [IDP] qualitative considerations are expected to adequately identify the safety-significant functions and SSCs required for those functions and no additional seismic reviews are necessary for 50.69 categorization.

The proposed categorization approach for VEGP 3&4 is a risk-informed graded approach that is demonstrated to produce categorization insights equivalent to a seismic PRA (SPRA). For Tier 1 plants, this approach relies on the insights gained from the SPRAs examined in EPRI 3002017583 along with confirmation that the site GMRS is low. The EPRI report demonstrates that seismic risk is adequately addressed for Tier 1 sites by the results of additional qualitative assessments discussed in this section and existing elements of the categorization process specified in NEI 00-04.

For example, the 10 CFR 50.69 categorization process as defined in NEI 00-04 includes an Integral Assessment that weighs the hazard-specific relative importance of a component (e.g., internal events, internal fire, seismic) by the fraction of the total Core Damage Frequency (CDF) contributed by that hazard. For Tier 1 sites, the seismic risk (CDF/Large Early Release Frequency (LERF)) will be low such that seismic hazard risk is unlikely to influence an HSS decision. In applying the EPRI 3002017583 process for Tier 1 sites to the VEGP 3&4 10 CFR 50.69 categorization process, the IDP will be provided with the rationale for applying the EPRI guidance and informed of plant SSC-specific seismic insights for their consideration in the HSS/LSS deliberations.

EPRI 3002017583 recommends a risk-informed graded approach for addressing the seismic hazard in the 10 CFR 50.69 categorization process. There are a number of seismic fragility fundamental concepts that support a graded approach and there are important characteristics

about the comparison of the seismic design basis (represented by the SSE) to the site-specific seismic hazard (represented by the GMRS) that support the selected thresholds between the three evaluation tiers in the EPRI report. The coupling of these concepts with the categorization process in NEI 00-04 are the key elements of the approach defined in EPRI 3002017583 for identifying unique seismic insights.

The seismic fragility of an SSC is a function of the margin between an SSC's seismic capacity and the site-specific seismic demand. References such as EPRI NP-6041 (Reference 10) provide inherent seismic capacities for most SSCs that are not directly related to the site-specific seismic demand. This inherent seismic capacity is based on the non-seismic design loads (pressure, thermal, dead weight, etc.) and the required functions for the SSC. For example, a pump has a relatively high inherent seismic capacity based on its design and that same seismic capacity applies at a site with a very low demand and at a site with a very high demand. At sites with lower seismic demands compared to the design basis (such as VEGP 3&4) there is no need to perform more detailed evaluations to demonstrate the inherent seismic capacities documented in industry sources such as EPRI NP-6041. Low seismic demand sites have lower likelihood of seismically-induced failures and lesser challenges to plant systems, providing the technical basis for allowing use of a graded approach for addressing seismic hazard at VEGP 3&4.

There are some plant features such as equipment anchorage that have seismic capacities more closely associated with the site-specific seismic demand since those specific features are specifically designed to meet that demand. However, even for these features, the design basis criteria have intended conservatisms that result in significant seismic margins within SSCs. These conservatisms are reflected in key aspects of the seismic design process. The SSCs used in nuclear power plants are intentionally designed using conservative methods and criteria to ensure that they have margins well above the required design bases. Experience has shown that design practices result in margins to realistic seismic capacities of 1.5 or more.

The following provides the basis for establishing Tier 1 criteria in EPRI 3002017583:

a. SSCs for which the inherent seismic capacities are applicable, or which are designed to the plant SSE, will have low probabilities of failure at sites where the peak spectral acceleration of the GMRS < 0.2g or where the GMRS < SSE between 1 and 10 Hz.

b. The low probabilities of failure of individual components would also apply to components considered to have correlated seismic failures.

c. These low probabilities of failure lead to low seismic CDF and LERF estimates, from an absolute risk perspective.

d. The low seismic CDF and LERF estimates lead to reasonable confidence that seismic risk contributions would allow reducing an HSS to LSS due to the 10 CFR 50.69 Integral Assessment provided the equipment is HSS only due to seismic considerations.

Test cases described in Section 3 of EPRI 3002017583 showed that it would be unusual, even for moderate hazard plants, to exhibit any unique seismic insights, including those resulting from correlated failures. While it is prudent to perform additional evaluations to identify conditions where correlated failures may occur for Tier 2 sites, for Tier 1 sites such as VEGP 3&4,

correlation studies would not lead to new seismic insights or affect the site risk results in any significant way.

The test case information from EPRI 3002017583, developed by other licensees, including Case Study A (Reference 11), Case Study C (Reference 12), and Case Study D (Reference 13), as well as responses to requests for additional information (RAIs) and amendments (References 14,15,16,17,18,19,20,21,22) that clarify aspects of these case studies, provide additional supporting bases for this application. Therefore, these case studies, RAI responses, and amendments are incorporated by reference into this amendment request.

The Tier 1 to Tier 2 threshold as defined in EPRI 3002017583 provides a clear and traceable boundary that can be consistently applied from plant site to plant site. Additionally, because the boundary is well defined, if new information is obtained on the site hazard, a site's location within a particular tier can be readily confirmed. In the unlikely event that the VEGP 3&4 seismic hazard changes to medium risk (i.e., Tier 2) at some future time, SNC will follow its categorization review and adjustment process procedures to review the changes to the plant and update, as appropriate, the SSC categorization in accordance with 10 CFR 50.69(e). In accordance with the proposed license condition, SNC will seek prior NRC approval if it is determined that a process different from the proposed alternative seismic Tier 1 approach is warranted for seismic risk consideration in categorization under 10 CFR 50.69 (e.g., a change from the Tier 1 approach to the Tier 2 approach).

The following provides the basis for concluding that VEGP 3&4 meets the Tier 1 site criteria.

On March 12, 2012, the NRC issued the 50.54(f) letter associated with post-Fukushima recommendations (Reference 23). In accordance with this letter, holders of COLs under 10 CFR Part 52 were not required to respond to Enclosures 1 through 4 because the issues addressed in Near-Term Task Force (NTTF) Recommendations 2.1 and 2.3 regarding seismic and flooding re-evaluations and walkdowns were resolved. SNC acknowledged that Enclosures 1 through 4 related to Recommendations 2.1 and 2.3 were not applicable to VEGP 3&4 (Reference 24).

Following SNC's submittal of the VEGP 1&2 Seismic Hazard and Screening Report (Reference 25), the NRC issued a letter to SNC on November 5, 2014, requesting an updated GMRS and Foundation Input Response Spectra (FIRS) for VEGP 3&4 (Reference 26). The NRC also requested that SNC describe whether the updated FIRS for VEGP 3&4 were bounded by the AP1000 Certified Seismic Design Response Spectra (CSDRS).

On December 5, 2014, SNC responded to the NRC (Reference 27) and described a reassessment of the impact on the seismic design margin for VEGP 3&4 considering the VEGP 1&2 2.1 site hazard provided in the 2.1-Seismic submittal (Reference 25). SNC's response provided the following summary of the results of the reassessment:

> … the VEGP 1&2 [GMRS] … did exceed the VEGP 3&4 GMRS … . As a result of the noted exceedance, SNC performed a re-assessment of the seismic response of the VEGP 3&4 Nuclear Island similar to that provided in the VEGP 3&4 Combined Operating License Application … based on using the GMRS from VEGP 1&2. The results of this re-assessment continues to support the conclusion provided in the COLA that the seismic demand used for the standard AP1000 design bounds the VEGP 3&4 site specific seismic demand and thus still maintains the significant design margin. Although the latest Central and Eastern United States (CEUS) methodology was used in this re-assessment, it is not

> the design basis of either VEGP 1&2 or VEGP 3&4. The in-structure response analysis that was performed did not provide any basis to change the design basis for VEGP 3&4.
>
> Therefore, it is concluded that the VEGP 1&2 site hazard provided in the 2.1 seismic submittal does not impact the conclusions regarding seismic as provided in the VEGP 3&4 COLA.

The NRC reviewed SNC's response and issued a letter dated August 12, 2015 (Reference 28). In its evaluation of SNC's reassessment of the impact on the seismic design margin for VEGP 3&4, the NRC stated the following:

> The staff agrees that the use of the VEGP Units 1 and 2 GMRS for VEGP Units 3 and 4 is appropriate given the proximity and similarity of subsurface conditions. Also, the licensee evaluated the impact on the seismic design margin for VEGP Units 3 and 4 due to the updated FIRS developed from the VEGP Units 1 and 2 GMRS. The licensee's evaluation followed the same analysis approach described in Vogtle Electric Generating Plant Units 3 and 4 COL Application Part 2 Final Safety Analysis Report, Revision 5, Section 3.7 Seismic Design Appendix 3G Nuclear Island Seismic Analysis. The licensee's December 5, 2014, submittal included figures showing the ISRS comparisons at the six key locations based on the updated FIRS. The staff reviewed these figures, which demonstrate that the updated in-structure seismic demands are bounded by the AP1000 standard design seismic demands. Based on the staff review, no change in the seismic licensing basis or further information is necessary.

The information in Reference 27 showed that the VEGP 3&4 ISRS were essentially enveloped by the AP1000 standard design ISRS, thereby demonstrating significant seismic design margin. The NRC's conclusion confirmed SNC's assertions that the in-structure seismic demands at VEGP 3&4 were bounded by the AP1000 standard design seismic demands.

Subsequent to the events at Fukushima Dai-ichi, the NRC implemented the Process for the Ongoing Assessment of Natural Hazards Information (POANHI) in accordance with the Staff Requirements Memorandum (SRM), SRM-SECY-16-0144 (Reference 29). On May 15, 2023, the NRC issued a letter to SNC to document the NRC's latest understanding of seismic hazards at the VEGP site, following the POANHI (Reference 30). The NRC staff assessment considered new seismic ground motion attenuation models for the CEUS and updated seismic site response methods. This assessment included a screening evaluation comparing the updated GMRS and seismic hazard curves developed by SNC for VEGP 1&2 in response to the 50.54(f) letter, and VEGP 3&4 for its Early Site Permit (ESP) application. Based on the screening comparison of the GMRS and an evaluation of the change in seismic risk, the NRC staff determined that no further regulatory evaluation or action to modify the plant seismic risk licensing basis for VEGP Units 1, 2, and 3 was warranted. Unit 4 was not included in the evaluation because it was under construction at the time. Figure 3-2 of this enclosure was recreated from Figure 10 of Reference 30, and shows the updated GMRS (red curve) compared to (1) the GMRS (black curve) developed for NUREG/KM-0017, (2) the GMRS (blue curve) in SNC's Seismic Hazard and Screening Report (SHSR) and SPRA assessment for VEGP 1&2, and (3) the GMRS (green curve) developed by SNC for its ESP.

The information from the NRC's May 15, 2023, POANHI letter was used to compare the updated GMRS at the VEGP 3&4 site to the plant's design basis. The CSDRS represents the seismic design basis for VEGP 3&4 as approved under 10 CFR Part 52. The updated GMRS data from Table 3 of Reference 30 and the VEGP 3&4 CSDRS are plotted in Figure 3-3 of this

enclosure. As shown in Figure 3-3, the updated GMRS is below the CSDRS from 1 Hz to 10 Hz. Because the GMRS remains below the plant design basis between 1.0 Hz and 10 Hz, the VEGP 3&4 site meets the criteria for Tier 1 as defined in EPRI 3002017583 (Reference 3). The rationale for classifying VEGP 3&4 as a Tier 1 site will be documented and presented to the IDP for each categorized system.

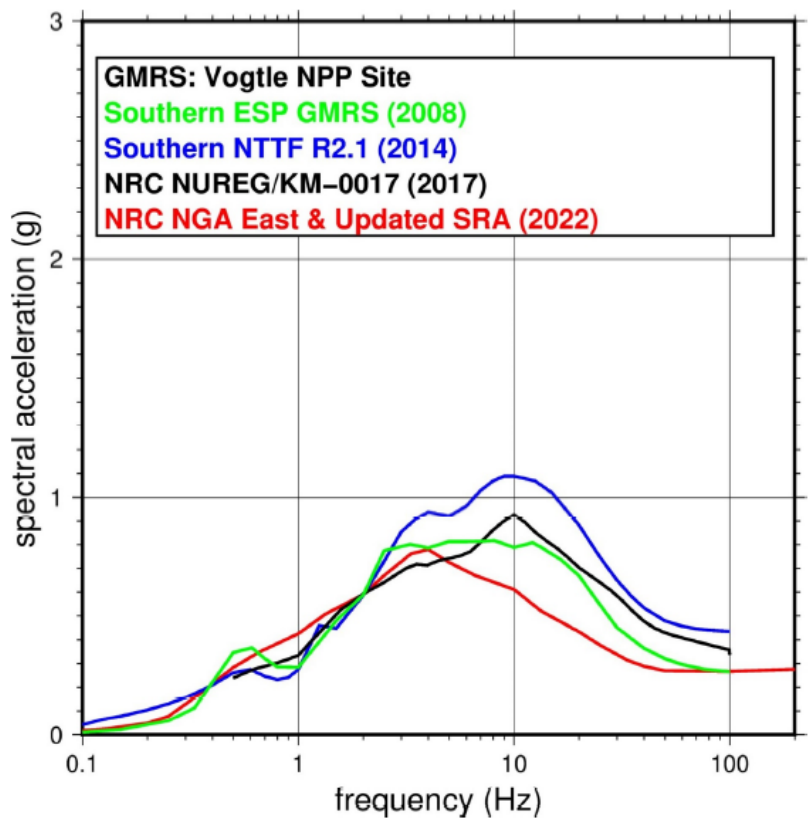

Figure 3-2: GMRS for the Vogtle Site

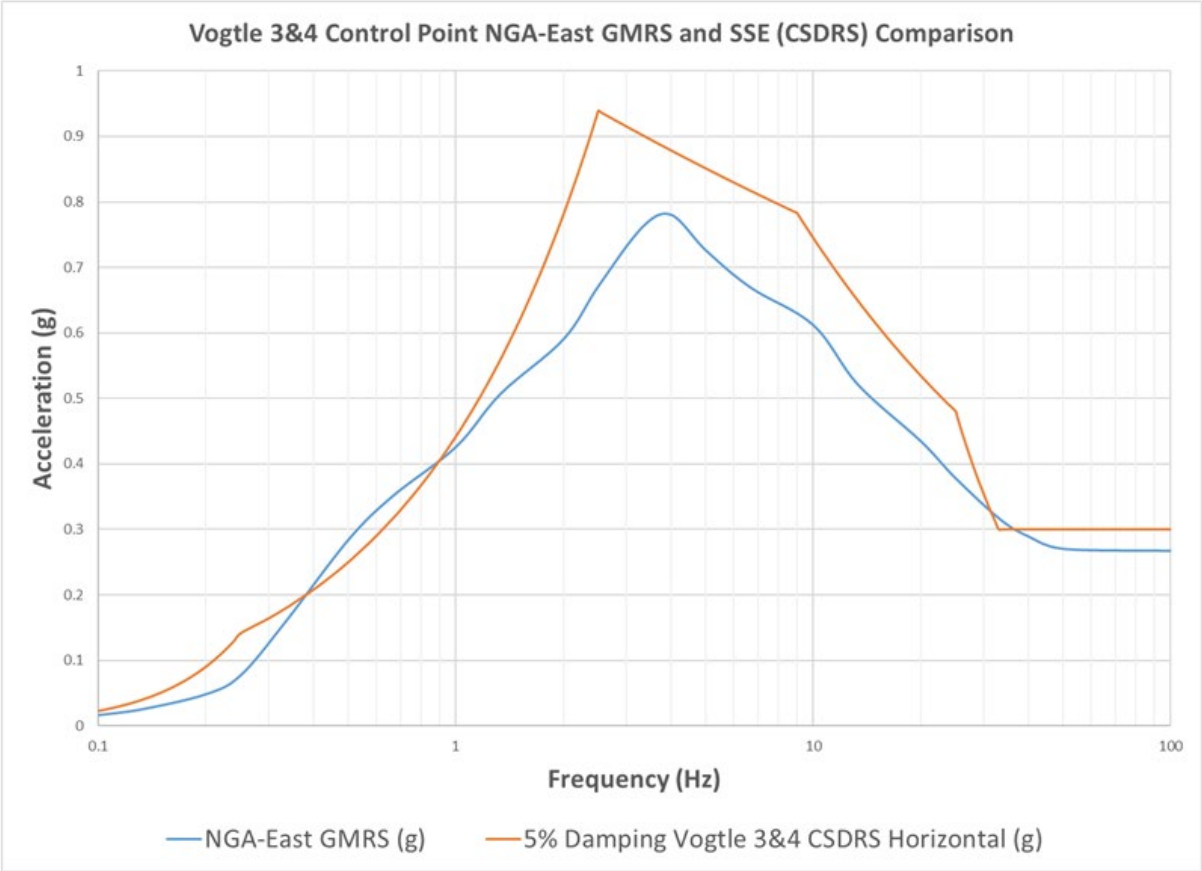

Figure 3-3: Vogtle Site Updated GMRS and CSDRS

For a Tier 1 site such as VEGP 3&4, the low seismic hazard relative to plant seismic capability makes it unlikely that any unique seismic condition would exist that would cause an SSC to be designated HSS. The best estimate seismic CDF (SCDF) and seismic LERF (SLERF) for VEGP 3&4 are 3.60E-07 and 3.35E-07, respectively, based on the seismic PRA. While the SCDF and SLERF contributions for VEGP 3&4 are perhaps not considered small when expressed as a percentage of the unit's total CDF and LERF, the absolute SCDF and SLERF values themselves are consistent, and in several cases lower than those reported for other plants approved for use of the EPRI approach for seismic Tier 1 sites. The percentage contribution for seismic risk at VEGP 3&4 is driven by the exceptionally low baseline CDF and LERF from internal events, internal flood, and fire for these AP1000 units. If the use of seismic PRA in the categorization process would produce any cases of a component that is HSS for seismic only, there is reasonable confidence the 10 CFR 50.69 Integral Assessment would allow an overall LSS result.

As an enhancement to the EPRI study results as they pertain to VEGP 3&4, the proposed categorization approach for seismic hazards will include qualitative consideration of the mitigation capabilities of SSCs during seismically induced events and seismic failure modes, based on insights obtained from prior seismic evaluations performed for VEGP 3&4. For example, the System Categorization Document (SCD) presented to the IDP will include a summary of identified plant-specific seismic insights pertinent to the system being categorized.

It will also state the basis for applicability of the EPRI 3002017583 study and the rationale for classifying VEGP 3&4 as a Tier 1 plant. The discussion of the Tier 1 bases will include such factors as:

- The low seismic hazard for the plant, which is subject to periodic reconsideration as new information becomes available through industry evaluations; and
- The definition of Tier 1 in the EPRI study.

At several steps of the categorization process (e.g., as noted in Figure 3-1 and Table 3-1) the categorization team will consider the available seismic insights relative to the system being categorized and document their conclusions in the SCD. Integrated importance measures over all the modeled hazards (i.e., internal events, internal flooding, and internal fire for VEGP 3&4) are calculated per Section 5.6 of NEI 00-04, and components for which these measures exceed the specified criteria are preliminary HSS which cannot be changed to LSS. HSS SSCs uniquely identified by the VEGP 3&4 PRA models also having design-basis functions during seismic events or functions credited for mitigation and prevention of severe accidents caused by seismic events will be addressed using non-PRA based qualitative assessments in conjunction with any seismic insights provided by the PRA.

For components that are HSS due to fire PRA but not HSS due to internal events PRA, the categorization team will review design-basis functions during seismic events or functions credited for mitigation and prevention of severe accidents caused by seismic events and characterize these for presentation to the IDP as additional qualitative inputs described in the SCD.

The categorization team will review available VEGP 3&4 plant-specific seismic reviews and other resources such as those identified above. The objective is to identify plant-specific seismic insights relevant to the components in the system being categorized, that might include potentially important impacts such as:

- Impact of relay chatter
- Implications related to potential seismic interactions such as with block walls
- Seismic failures of passive SSCs such as tanks and heat exchangers
- Any known structural or anchorage issues with a particular SSC
- Components implicitly part of PRA-modeled functions (including relays)
- Components that may be subject to correlated failures

These impacts would be compiled on an SSC basis. As each system is categorized, the system-specific seismic insights will be provided to the IDP as part of the review process. The IDP can challenge, from a seismic perspective, any candidate LSS recommendation for any SSC if they believe there is a basis for doing so. Any decision by the IDP to downgrade preliminary HSS components to LSS will also consider the applicable seismic insights in that decision. These insights will provide the IDP a means to consider potential impacts of seismic events in the categorization process.

Use of the EPRI approach outlined in Reference 3 to assess seismic hazard risk for 10 CFR 50.69 with the additional reviews discussed above will provide a process for categorization of RISC-1, RISC-2, RISC-3, and RISC-4 SSCs that satisfies the requirements of 10 CFR 50.69(c).

Based on the above, the Summary/Conclusion/Recommendation from Section 2.2.3 of Reference 3 applies to VEGP 3&4, i.e., VEGP 3&4 is a Tier 1 plant for which the plant design bounds the GMRS; therefore, unique seismic categorization insights are expected to be minimal. As discussed in Reference 3, the likelihood of identifying a unique seismic insight that would cause an SSC to be designated HSS is very low. Calvert Cliff's responses to RAIs (References 31, 32, and 33) are incorporated by reference into this amendment request as they provide additional supporting bases for Tier 1 plants and apply to VEGP 3&4. Therefore, with little to no anticipated unique seismic insights, the 10 CFR 50.69 categorization process using the FPIE PRA and other risk evaluations, along with the IDP's DID and qualitative assessment, effectively identifies the safety-significant functions and SSCs.

3.2.4 Other External Hazards

All other external hazards were screened for applicability to VEGP 3&4. A systematic process was used to identify all potential external hazards that could affect the site, including both natural and man-made hazards, as well as site-specific and plant-unique hazards. The review incorporated previously examined hazard information from the VEGP 1&2 IPEEE and the VEGP 3&4 UFSAR. Attachment 4 of this enclosure provides a summary of the screening results for other external hazards, while Attachment 5 outlines the progressive screening approach.

3.2.5 Low Power and Shutdown

Consistent with NEI 00-04, the SNC categorization process will use the shutdown safety management plan described in NUMARC 91-06 (Reference 7) for evaluation of safety significance related to low power and shutdown conditions. The overall process for addressing shutdown risk is illustrated in Figure 5-7 of NEI 00-04.

NUMARC 91-06 specifies that a DID approach should be used with respect to each defined shutdown key safety function. The key safety functions defined in NUMARC 91-06 are evaluated for categorization of SSCs.

SSCs that meet either of the two criteria (i.e., considered part of a "primary shutdown safety system" or a failure would initiate an event during shutdown conditions) described in Section 5.5 of NEI 00-04 will be considered preliminary HSS.

3.2.6 PRA Maintenance and Updates

The SNC risk management process ensures that the applicable PRA models used in this application continue to reflect the as-built and as-operated plant. The process delineates the responsibilities and guidelines for updating the PRA models and includes criteria for both regularly scheduled and interim PRA model updates. The process includes provisions for monitoring potential areas affecting the PRA models (e.g., due to changes in the plant, errors or limitations identified in the model, and industry operational experience) for assessing the risk impact of unincorporated changes, and for controlling the model and associated computer files. The process will assess the impact of these changes on the plant PRA model in a timely manner but no longer than once every two refueling outages. If there is a significant impact on the PRA model, the SSC categorization will be reevaluated.

In addition, SNC will implement a process that addresses the requirements in NEI 00-04, Section 11, "Program Documentation and Change Control." The process will review the results of periodic and interim updates of the plant PRA that may affect the results of the categorization

process. If the results are affected, adjustments will be made as necessary to the categorization or treatment processes to maintain the validity of the processes. In addition, any PRA model upgrades will be peer reviewed prior to implementing those changes in the PRA model used for categorization.

3.2.7 PRA Uncertainty Evaluations

Uncertainty evaluations associated with any applicable baseline PRA model(s) used in this application were evaluated during the assessment of PRA technical adequacy and confirmed through the self-assessment and peer review processes as discussed in Section 3.3 of this enclosure.

Uncertainty evaluations associated with the risk categorization process are addressed using the processes discussed in Section 8 of NEI 00-04 and in the prescribed sensitivity studies discussed in Section 5.

In the overall risk sensitivity studies, SNC will utilize a factor of 3 to increase the unavailability or unreliability of LSS components consistent with that approved for VEGP 1&2 in Reference 4. Consistent with the NEI 00-04 guidance, SNC will perform both an initial sensitivity study and a cumulative sensitivity study. The initial sensitivity study applies to the system that is being categorized. In the cumulative sensitivity study, the failure probabilities (unreliability and unavailability, as appropriate) of all LSS components modeled in all identified PRA models for all systems that have been categorized are increased by a factor of 3. This sensitivity study together with the periodic review process assures that the potential cumulative risk increase from the categorization is maintained acceptably low. The performance monitoring process monitors component performance to ensure that potential increases in failure rates of categorized components are detected and addressed before reaching the rate assumed in the sensitivity study.

The VEGP 3&4 internal events and fire PRA models and documentation were reviewed for plant-specific and generic modeling assumptions and related sources of uncertainty. The process to evaluate uncertainties is defined in NUREG-1855 (Reference 34) and EPRI Technical Report TR-1016737 (Reference 35). Each PRA model includes an evaluation of the potential sources of uncertainty for the base case models using the approach that is consistent with the ASME/ANS RA-Sa-2009 requirements for identification and characterization of uncertainties and assumptions.

Each PRA element notebook was also reviewed for assumptions and sources of uncertainties. The characterization of assumptions and sources of uncertainties are based on whether the assumption and/or source of uncertainty is key to the 50.69 application in accordance with RG 1.200 (References 36 and 37). Key VEGP 3&4 PRA model specific assumptions and sources of uncertainty for this application were identified and dispositioned in Attachment 6. The conclusion of this review is that no additional sensitivity analyses are required to address VEGP 3&4 PRA model specific assumptions or sources of uncertainty.

3.3 PRA Review Process Results (10 CFR 50.69(b)(2)(iii))

The PRA models described in Section 3.2 were initially assessed against RG 1.200, "An Approach for Determining the Technical Adequacy of Probabilistic Risk Assessment Results for Risk-Informed Activities," Revision 2 (Reference 36). Follow on peer reviews assessed the PRA models against RG 1.200, Revision 3 (Reference 37).

Attachment 3 provides a summary of the peer review history, including remaining open findings and their impact on this application. This information demonstrates that the VEGP 3&4 PRA models are of sufficient quality and level of detail to support the categorization process. Furthermore, each model has undergone a peer review process evaluated against a standard or set of acceptance criteria endorsed by the NRC, as required by 10 CFR 50.69(c)(1)(i).

3.4 Risk Evaluations (10 CFR 50.69(b)(2)(iv))

The SNC 10 CFR 50.69 categorization process will implement the guidance in NEI 00-04. The overall risk evaluation process described in the NEI guidance addresses both known degradation mechanisms and common cause interactions and meets the requirements of § 50.69(b)(2)(iv). Sensitivity studies described in NEI 00-04, Section 8, will be used to confirm that the categorization process results in acceptably small increases to CDF and LERF. The failure rates for equipment and initiating event frequencies used in the PRA include the quantifiable impacts from known degradation mechanisms, as well as other mechanisms (e.g., design errors, manufacturing deficiencies, and human errors). Subsequent performance monitoring and PRA updates required by the rule will continue to capture this data and provide timely insights into the need to account for any important new degradation mechanisms.

3.5 Feedback and Adjustment Process

If significant changes to the plant risk profile are identified, or if it is identified that a RISC-3 or RISC-4 SSC can (or actually did) prevent a safety significant function from being satisfied, a timely evaluation and review will be performed prior to the normally scheduled periodic review. Otherwise, the assessment of potential equipment performance changes and new technical information will be performed during the normally scheduled periodic review cycle.

To more specifically address the feedback and adjustment process (i.e., performance monitoring) as it pertains to the proposed VEGP 3&4 Tier 1 approach discussed in Section 3.2.3 of this enclosure, implementation of the SNC design control and corrective action programs provide assurance that the inputs for the qualitative determinations for seismic continue to remain valid to maintain compliance with the requirements of 10 CFR 50.69(e).

The performance monitoring process is described in SNC's 10 CFR 50.69 program documents. The program requires that the periodic review assess changes that could impact the categorization results and provides the IDP with an opportunity to recommend categorization and treatment adjustments. Personnel from engineering, operations, risk-informed engineering, regulatory affairs, and others have responsibilities for preparing and conducting various performance monitoring tasks that feed into this process. The intent of the performance monitoring reviews is to discover trends in component reliability; to identify and reverse negative performance trends and take corrective action, if necessary.

The SNC configuration control process ensures that changes to the plant, including a physical change to the plant and/or changes to documents, are evaluated to determine the impact to drawings, design bases, licensing documents, programs, procedures, and training.

SNC has a comprehensive problem identification and corrective action program that ensures that issues are identified and resolved. Issues that may impact the 10 CFR 50.69 categorization process will be identified and addressed through the problem identification and corrective action program, including seismic-related issues.

The SNC 10 CFR 50.69 program requires that system categorization cannot be approved by the IDP until the panel's comments have been resolved to the satisfaction of the IDP. This includes issues related to system-specific seismic insights considered by the IDP during categorization.

Scheduled periodic reviews are completed at least once every two refueling cycles and will evaluate new insights resulting from available risk information (i.e., PRA model or other analysis used in the categorization) changes, design changes, operational changes, and SSC performance. If it is determined that these changes have affected the risk information or other elements of the categorization process such that the categorization results are more than minimally affected, then the risk information and the categorization process will be updated. This scheduled review will include:

- A review of plant modifications since the last review that could impact the SSC categorization.
- A review of plant specific operating experience that could impact the SSC categorization.
- A review of the impact of the updated risk information on the categorization process results.
- A review of the importance measures used for screening in the categorization process.
- An update of the risk sensitivity study performed for the categorization.

In addition to the normally scheduled periodic reviews, if a PRA model or other risk information is upgraded, a review of the SSC categorization will be performed.

The periodic monitoring requirements of the 10 CFR 50.69 process will capture these issues and address them at a frequency commensurate with the issue severity. The 10 CFR 50.69 periodic monitoring program includes immediate and periodic reviews, that include the requirements of the regulation, to provide assurance that issues that could affect 10 CFR 50.69 categorization are addressed. The periodic monitoring process also monitors the performance and condition of categorized SSCs such that the assumptions for reliability in the categorization process are maintained.

4. REGULATORY EVALUATION

4.1 Applicable Regulatory Requirements/Criteria

The following NRC requirements and guidance documents are applicable to the proposed change.

- The regulations in Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50.69, "Risk-informed categorization and treatment of structures, systems and components for nuclear power reactors"
- NRC Regulatory Guide 1.201, "Guidelines for Categorizing Structures, Systems, and Components in Nuclear Power Plants According to their Safety Significance," Revision 1, May 2006

- Regulatory Guide 1.174, "An Approach for Using Probabilistic Risk Assessment in Risk-Informed Decisions on Plant-Specific Changes to the Licensing Basis," Revision 3, January 2018
- Regulatory Guide 1.200, "An Approach for Determining the Technical Adequacy of Probabilistic Risk Assessment Results for Risk-Informed Activities," Revision 2, March 2009
- Regulatory Guide 1.200, "Acceptability of Probabilistic Risk Assessment Results for Risk-Informed Activities," Revision 3, December 2020

The proposed change is consistent with the applicable regulations and regulatory guidance.

4.2 No Significant Hazards Consideration Analysis

Southern Nuclear Operating Company (SNC) proposes to modify the licensing basis for Vogtle Electric Generating Plant, Units 3 and 4 (VEGP 3&4), to allow for the voluntary implementation of the provisions of Title 10 of the *Code of Federal Regulations* (10 CFR), Part 50.69, "Risk-informed categorization and treatment of structures, systems and components for nuclear power reactors." The provisions of 10 CFR 50.69 allow adjustment of the scope of equipment subject to special treatment controls (e.g., quality assurance, testing, inspection, condition monitoring, assessment, and evaluation). For equipment determined to be of low safety significance, alternative treatment requirements can be implemented in accordance with this regulation. For equipment determined to be of high safety significance, requirements will not be changed or will be enhanced. This allows improved focus on equipment that has greater safety significance resulting in improved plant safety.

SNC has evaluated whether or not a significant hazards consideration is involved with the proposed amendments by focusing on the three standards set forth in 10 CFR 50.92, "Issuance of amendment," as discussed below:

1. Does the proposed change involve a significant increase in the probability or consequences of an accident previously evaluated?

   Response: No.

   The proposed change will permit the use of a risk-informed categorization process to modify the scope of structures, systems and components (SSCs) subject to NRC special treatment requirements and to implement alternative treatments per the regulations. The process used to evaluate SSCs for changes to NRC special treatment requirements and the use of alternative requirements ensures the ability of the SSCs to perform their design function. The potential change to special treatment requirements does not change the design and operation of the SSCs. As a result, the proposed change does not significantly affect any initiators to accidents previously evaluated or the ability to mitigate any accidents previously evaluated. The consequences of the accidents previously evaluated are not affected because the mitigation functions performed by the SSCs assumed in the safety analysis are not being modified. The SSCs required to safely shut down the reactor and maintain it in a safe shutdown condition following an accident will continue to perform their design functions.

   Therefore, the proposed change does not involve a significant increase in the probability or consequences of an accident previously evaluated.

2. Does the proposed change create the possibility of a new or different kind of accident from any accident previously evaluated?

   Response: No.

   The proposed change will permit the use of a risk-informed categorization process to modify the scope of SSCs subject to NRC special treatment requirements and to implement alternative treatments per the regulations. The proposed change does not change the functional requirements, configuration, or method of operation of any SSC. Under the proposed change, no additional plant equipment will be installed.

   Therefore, the proposed change does not create the possibility of a new or different kind of accident from any accident previously evaluated.

3. Does the proposed change involve a significant reduction in a margin of safety?

   Response: No.

   The proposed change will permit the use of a risk-informed categorization process to modify the scope of SSCs subject to NRC special treatment requirements and to implement alternative treatments per the regulations. The proposed change does not affect any Safety Limits or operating parameters used to establish the safety margin. The safety margins included in analyses of accidents are not affected by the proposed change. The regulation requires that there be no significant effect on plant risk due to any change to the special treatment requirements for SSCs and that the SSCs continue to be capable of performing their design basis functions, as well as to perform any beyond design basis functions consistent with the categorization process and results.

   Therefore, the proposed change does not involve a significant reduction in a margin of safety.

Based on the above, SNC concludes that the proposed change presents no significant hazards consideration under the standards set forth in 10 CFR 50.92(c), and accordingly, a finding of "no significant hazards consideration" is justified.

4.3 Conclusions

In conclusion, based on the considerations discussed above, (1) there is reasonable assurance that the health and safety of the public will not be endangered by operation in the proposed manner, (2) such activities will be conducted in compliance with the Commission's regulations, and (3) the issuance of the amendment will not be inimical to the common defense and security or to the health and safety of the public.

5. ENVIRONMENTAL CONSIDERATION

A review has determined that the proposed amendment would change a requirement with respect to installation or use of a facility component located within the restricted area, as defined in 10 CFR 20, or would change an inspection or surveillance requirement. However, the proposed amendment does not involve (i) a significant hazards consideration, (ii) a significant change in the types or a significant increase in the amounts of any effluents that may be released offsite, or (iii) a significant increase in individual or cumulative occupational radiation

exposure. Accordingly, the proposed amendment meets the eligibility criterion for categorical exclusion set forth in 10 CFR 51.22(c)(9). Therefore, pursuant to 10 CFR 51.22(b), no environmental impact statement or environmental assessment need be prepared in connection with the proposed amendment.

6. REFERENCES

1. Nuclear Energy Institute (NEI) 00-04, "10 CFR 50.69 SSC Categorization Guideline," Revision 0, July 2005 (ML052910035)
2. NRC Regulatory Guide 1.201, "Guidelines for Categorizing Structures, Systems, and Components in Nuclear Power Plants According to their Safety Significance," Revision 1, May 2006 (ML061090627)
3. Electric Power Research Institute (EPRI) 3002017583, "Alternative Approaches for Addressing Seismic Risk in 10 CFR 50.69 Risk-Informed Categorization," Technical Update, February 2020 (ML21082A170)
4. NRC letter to SNC, "Vogtle Electric Generating Plant, Units 1 and 2 – Issuance of Amendments Re: Use of 10 CFR 50.69," December 17, 2014 (ML14237A034)
5. SNC Letter to NRC, "Vogtle Electric Generating Plant Units 3 and 4, License Amendment Request to Revise Technical Specifications to Include Risk Informed Completion Times," January 29, 2026 (ML26029A439)
6. NRC letter to SNC, "Vogtle Electric Generating Plant, Units 3 and 4 – Exemption to 10 CFR 50.69, 'Risk-Informed Categorization and Treatment of Structures, Systems and Components for Nuclear Power Reactors'," February 10, 2026 (ML25262A038, ML25262A048, ML25262A049)
7. NUMARC 91-06, "Guidelines for Industry Actions to Assess Shutdown Management," December 1991 (ML14365A203)
8. NRC Letter to Entergy, "Arkansas Nuclear One, Unit 2 - Approval of Request for Alternative ANO2-R&R-004, Revision 1, Request to Use Risk-Informed Safety Classification and Treatment for Repair/Replacement Activities in Class 2 and 3 Moderate and High Energy Systems," April 22, 2009 (ML090930246)
9. NRC Regulatory Guide 1.147, "Inservice Inspection Code Case Acceptability, ASME Section XI, Division 1," Revision 15, October 2007 (ML072070419)
10. Electric Power Research Institute (EPRI) NP-6041-SL, "A Methodology for Assessment of Nuclear Power Plant Seismic Margin, Revision 1," August 1991
11. Exelon (Peach Bottom) Letter to NRC, "Supplemental Information to Support Application to Adopt 10 CFR 50.69, 'Risk-informed categorization and treatment of structures, systems, and components for nuclear power plants'," June 6, 2018 (ML18157A260)
12. SNC Letter to NRC, "Vogtle Electric Generating Plant – Units 1&2 License Amendment Request to Incorporate Seismic Probabilistic Risk Assessment into the 10 CFR 50.69 Categorization Process Response to Request for Additional Information (RAIs 4-11)," February 21, 2018 (ML18052B342)

13. TVA Letter to NRC, "Watts Bar Nuclear Plant, Units 1 and 2, Application to Adopt 10 CFR 50.69, 'Risk-informed Categorization and Treatment of Structures, Systems, and Components for Nuclear Power Reactors'," November 29, 2018 (ML18334A363)
14. Exelon (Peach Bottom) Letter to NRC, "Seismic Probabilistic Risk Assessment Report, Response to NRC Request for Information Pursuant to 10 CFR 50.54(f) Regarding Recommendation 2.1 of the Near-Term Task Force Review of Insights from the Fukushima Dai-ichi Accident," August 28, 2018 (ML18240A065)
15. NRC Letter to Exelon, "Peach Bottom Atomic Power Station, Units 2 and 3 – Staff Review of Seismic Probabilistic Risk Assessment Associated With Reevaluated Seismic Hazard Implementation of the Near-Term Task Force Recommendation 2.1: Seismic," June 10, 2019 (ML19053A469)
16. NRC Letter to Exelon, "Peach Bottom Atomic Power Station, Units 2 and 3 – Correction Regarding Staff Review of Seismic Probabilistic Risk Assessment Associated with Reevaluated Seismic Hazard Implementation of the Near-Term Task Force Recommendation 2.1: Seismic," October 8, 2019 (ML19248C756)
17. SNC Letter to NRC, "Vogtle Electric Generating Plant – Units 1 and 2 License Amendment Request to Modify Approved 10 CFR 50.69 Categorization Process," June 22, 2017 (ML17173A875)
18. NRC Letter to SNC, "Vogtle Electric Generating Plant, Units 1 and 2 – Issuance of Amendments Regarding Application of Seismic Probabilistic Risk Assessment into the Previously Approved 10 CFR 50.69 Categorization Process," August 10, 2018 (ML18180A062)
19. TVA Letter to NRC, "Seismic Probabilistic Risk Assessment for Watts Bar Nuclear Plant, Units 1 and 2 - Response to NRC Request for Information Pursuant to Title 10 of the Code of Federal Regulations 50.54(f) Regarding Recommendation 2.1 of the Near-Term Task Force Review of Insights from the Fukushima Dai-ichi Accident," June 30, 2017 (ML17181A485)
20. TVA Letter to NRC, "Tennessee Valley Authority (TVA) - Watts Bar Nuclear Plant Seismic Probabilistic Risk Assessment Supplemental Information," April 10, 2018 (ML18100A966)
21. NRC Letter to TVA, "Watts Bar Nuclear Plant, Units 1 and 2 – Staff Review of Seismic Probabilistic Risk Assessment Associated with Reevaluated Seismic Hazard Implementation of the Near-Term Task Force Recommendation 2.1: Seismic," July 10, 2018 (ML18115A138)
22. NRC letter to TVA, "Watts Bar Nuclear Plant, Units 1 and 2 - Issuance of Amendment Nos. 134 and 38 Regarding Adoption of Title 10 of the Code of Federal Regulations Section 50.69, 'Risk-Informed Categorization and Treatment of Structures, Systems, and Components for Nuclear Power Plants'," April 30, 2020 (ML20076A194)
23. NRC Letter to all Power Reactor Licensees, "Request for Information Pursuant to Title 10 of the Code of Federal Regulations 50.54(f) Regarding Recommendations 2.1, 2.3, and 9.3, of the Near-Term Task Force Review of Insights from the Fukushima Dai-ichi Accident," March 12, 2012 (ML12053A340)
24. SNC Letter to NRC, "Vogtle Electric Generating Plant Units 3 and 4 30 Day Response to March 12, 2012 Information Request," April 4, 2012 (ML12097A052)

25. SNC Letter to NRC, "Vogtle Electric Generating Plant – Units 1 and 2 Seismic Hazard and Screening Report for CEUS Sites," March 31, 2014 (ML14092A019)
26. NRC Letter to SNC, "Request for Submittal of Updated Ground Motion Spectra and Foundation Input Response Spectra," November 5, 2014 (ML14302A180)
27. SNC to NRC, "Vogtle Electric Generation Plant – Units 3 and 4 Response to Request for Submittal of Updated Ground Motion Spectra and Foundation Input Response Spectra," December 5, 2014 (ML14339A849)
28. NRC Letter to SNC, "Vogtle Electric Generating Plant Units 3 and 4 – Updated Ground Motion Response Spectra," August 12, 2015 (ML15139A516)
29. NRC SRM, "SECY-15-0137 – Proposed Plans for Resolving Open Fukushima Tier 2 & 3 Recommendations," February 8, 2016 (ML16039A175)
30. NRC Letter to SNC, "Staff Assessment of Updated Seismic Hazard Information and Latest Understanding of Seismic Hazards at the Vogtle Plant Site Following the NRC Process for the Ongoing Assessment of Natural Hazards Information," May 15, 2023 (ML23006A091)
31. Exelon (Calvert Cliffs) Letter to NRC, "Response to Request for Additional Information Regarding the Application to Adopt 10 CFR 50.69, 'Risk-informed categorization and treatment of structures, systems, and components for nuclear power reactors'," July 1, 2019 (ML19183A012)
32. Exelon (Calvert Cliffs) Letter to NRC, "Response to Request for Additional Information Regarding the Application to Adopt 10 CFR 50.69, 'Risk-informed categorization and treatment of structures, systems, and components for nuclear power reactors'," July 19, 2019 (ML19200A216)
33. Exelon (Calvert Cliffs) Letter to NRC, "Revised Response to Request for Additional Information Regarding the Application to Adopt 10 CFR 50.69, 'Risk-Informed Categorization and Treatment of Structures, Systems, and Components for Nuclear Power Reactors'," August 5, 2019 (ML19217A143)
34. NUREG-1855, "Guidance on the Treatment of Uncertainties Associated with PRAs in Risk-Informed Decisionmaking," Revision 1, March 2017 (ML17062A466)
35. EPRI TR-1016737, "Treatment of Parameter and Model Uncertainty for Probabilistic Risk Assessments," December 2008
36. Regulatory Guide 1.200, "An Approach for Determining the Technical Adequacy of Probabilistic Risk Assessment Results for Risk-Informed Activities," Revision 2, March 2009 (ML090410014)
37. Regulatory Guide 1.200, "Acceptability of Probabilistic Risk Assessment Results for Risk-Informed Activities," Revision 3, December 2020 (ML20238B871)

Attachment 1: List of Categorization Prerequisites

SNC has established fleet procedures that outline the process for performing 10 CFR 50.69 categorization. These fleet procedures will be implemented at Vogtle Electric Generating Plant, Units 3 and 4 (VEGP 3&4) prior to performing system categorizations on VEGP 3&4 systems. The fleet procedures to be implemented at VEGP 3&4 contain the elements/steps listed below.

- Integrated Decision-Making Panel (IDP) member qualification requirements.
- Qualitative assessment of system functions. System functions are qualitatively categorized as preliminary High Safety Significant (HSS) or Low Safety Significant (LSS) based on the seven criteria in Section 9 of NEI 00-04. Any component supporting an HSS function is categorized as preliminary HSS. Components supporting an LSS function are categorized as preliminary LSS.
- Component safety significance assessment. Safety significance of active components is assessed through a combination of Probabilistic Risk Assessment (PRA) and non-PRA methods, covering all hazards. Safety significance of passive components is assessed using a methodology for passive components.
- Assessment of defense-in-depth (DID) and safety margin. Safety-related components that are categorized as preliminary LSS are evaluated for their role in providing DID and safety margin and, if appropriate, upgraded to HSS.
- Review by the IDP. The categorization results are presented to the IDP for review and approval. The IDP reviews the categorization results and makes the final determination on the safety significance of system functions and components.
- Risk sensitivity study. For PRA-modeled components, an overall risk sensitivity study is used to confirm that the population of preliminary LSS components results in acceptably small increases to core damage frequency (CDF) and large early release frequency (LERF) and meets the acceptance guidelines of Regulatory Guide 1.174.
- Periodic reviews are performed to ensure continued categorization validity and acceptable performance for those SSCs that have been categorized.
- Documentation requirements per Section 3.1.1 of the enclosure.

Attachment 2: Description of PRA Models Used in Categorization

Unit	Model	Baseline CDF (/yr)	Baseline LERF (/yr)	Comment
3	Internal Events	3.59E-07	3.06E-08	The PRA models used to support categorization under 10 CFR 50.69 are the same models that are being reviewed by the NRC as part of adopting Technical Specifications Task Force (TSTF)-505, Revision 2.
3	Internal Flooding	1.98E-07	5.05E-08	
3	Internal Fire	7.37E-07	2.33E-07	
4	Internal Events	3.59E-07	3.06E-08	
4	Internal Flooding	1.98E-07	5.05E-08	
4	Internal Fire	7.38E-07	2.32E-07	

Attachment 3: PRA Model Peer Review History and Disposition and Resolution of Open Peer Review Findings and Self-Assessment Open Items

1.0 Internal Events PRA Model

The Vogtle Electric Generating Plant, Units 3 and 4 (VEGP 3&4) internal events PRA model is an at-power model (i.e., it directly addresses plant configurations during plant modes 1 and 2 of reactor operation). The model includes both at-power internal events core damage frequency (CDF) and large early release frequency (LERF). Note that this portion of the VEGP 3&4 PRA model does not incorporate the risk impacts of external events. The treatment of seismic risk and other external hazards for this application are discussed in Attachment 4 and Section 3.2.3 of the enclosure.

1.1 RG 1.200 Previous and Current Peer Reviews – Internal Events PRA

The At-Power Internal Events PRA was peer reviewed against Part 2 of ASME/ANS RA-Sa-2009 in 2013. Since the 2013 peer review, the VEGP 3&4 plant At-Power Internal Events PRA was updated to address several peer review facts and observations (F&Os), to the extent possible for an early operational plant. Follow on work addressed further model improvement focused on as-built, as-operated plant information.

The initial peer review conducted in 2013 recognized certain limitations in meeting supporting requirements of the standard based on the model's pre-operational status, and the team also recognized key aspects of the plant's design. Table A3-1 identifies AP1000 design features along with potential impacts on the risk profile.

Table A3-1 AP1000 Design Features

Design Features	Potential Impact
Passive systems, including Passive Decay Heat Removal, Passive Core Cooling, and Passive Containment Cooling systems	Designed capability to prevent/mitigate severe accident scenarios with minimal active equipment and human interactions
Passive systems are fail-safe for up to 72 hours with loss of AC power	Increased Reliability; Improved Capability for station blackout (SBO) events
Canned reactor coolant pumps (RCPs)	Eliminates Potential for RCP Seal LOCA

Subsequently, the VEGP 3&4 Internal Events PRA Model was subject to a closure and focused-scope industry peer review conducted by the Pressurized Water Reactor Owners' Group (PWROG) following self-assessment in 2025. The peer review and subsequent independent assessments demonstrated the VEGP 3&4 Internal Events PRA substantially met the ASME/ANS PRA standard RA-Sa-2009.

1.2 Resolution of F&Os from RG 1.200 Internal Events Peer Review

The VEGP 3&4 PRA Internal Events Model was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted in 2025, with a number of open findings

remaining as either out of scope (not applicable) or not fully resolved to meet Category II or better. The findings that remain open are summarized in Table A3-2 along with the associated dispositions relative to the 10 CFR 50.69 program. An independent F&O closure assessment was conducted and the result are documented and available for audit.

Table A3-2 Summary of Peer Review Findings – Internal Events PRA			
F&O No.	**F&O Details**	**Associated Supporting Requirement (SR)**	**Impact on Application**
IE-A3-01	A number of SRs were not achievable in the IE element because insufficient plant-specific data are available for update.	IE-A3 – SR MET IE-A7 – SR MET IE-A9 – SR MET (CC II) IE-C2 – Not Met IE-C4 – Not Met	There is no significant impact to the 50.69 application. Generic Initiating event data was utilized. Update to include plant specific operating experience is not expected to result in significant deviations from generic industry data or significantly alter the risk profile.
DA-C2-01	Site specific guidance establishes methodologies for performing a data update. Due to the early operational state of the plant, sufficient operating experience is not yet available to support this task.	DA-C2 – Not Met DA-C3 – Not Met DA-C4 – Not Met DA-C5 – Not Met DA-C6 – Not Met DA-C7 – Not Met DA-C8 – Not Met DA-C9 – Not Met DA-C11 – Not Met DA-C13 – Not Met DA-C14 – Not Met DA-D1 – SR MET (CC I) DA-D4 – SR MET (CC I)	There is no significant impact to the 50.69 application. Generic availability and reliability data were utilized. Update to include plant specific operating experience is not expected to result in significant deviations from generic industry data or significantly alter the risk profile. Additionally, parametric uncertainty is adequately addressed by the prescribed sensitivity studies discussed in NEI 00-04.
QU-D4-01	No similar plant exists for comparison.	QU-D4 – SR MET (CC I)	There is no significant impact to the 50.69 application. Review of PRAs from similar plants, if available, would allow for comparison of risk profiles.
LE-E2-01	Model results show that LERF is dominated by two phenomena of 1) hydrogen	LE-D1 – SR MET (CC I) LE-E2 – SR MET (CC I)	There is no significant impact to the 50.69 application. While there is potential that these

Table A3-2 Summary of Peer Review Findings – Internal Events PRA			
F&O No.	**F&O Details**	**Associated Supporting Requirement (SR)**	**Impact on Application**
	Deflagration-to-Detonation Transition, and 2) diffusion flame. This SR is characterized as CC I because of the conservative nature of the assumptions leading to early containment failure.		systems, if categorized, may be overestimated in their impact to LERF, this is a conservative approach. Additionally, the holistic process will ensure that other systems contributions is adequately considered.

2.0 Internal Flooding PRA Model

The current internal flood model was integrated into the PRA model of record to quantify flood-induced scenarios to determine their importance to plant risk. As part of quantification, the effects of flood-induced failures were combined with the effects of other types of equipment failures or unavailability. This includes the effects caused by flooding and failures coincident with the flooding due to maintenance unavailability, independent equipment faults and other credible causes. The direct and indirect effects of the flood were included in the quantification.

2.1 RG 1.200 Previous and Current Peer Reviews – Internal Flooding PRA Model

The At-Power Internal Flooding PRA was initially peer reviewed against Part 3 of ASME/ANS RA-Sa-2009 in 2013. Since the 2013 peer review, the VEGP 3&4 plant At-Power Internal Flooding PRA was updated to address peer review F&Os, to the extent possible for an early operational plant. Follow on work addressed further model improvement focused on as-built, as-operated plant information. It should be noted that certain supporting requirements were deemed not applicable because the AP1000 plant design is a modular single unit plant; for example, there is no need to include shared areas, systems or structures in the plant partitioning task. An additional focused scope review was performed in 2020.

Following further F&O closure efforts, model maintenance, and self-assessment in September 2025, the VEGP 3&4 Internal Flooding PRA Model was subject to a focused-scope industry peer review conducted by the PWROG that included a closure review to address previous findings. The peer review and subsequent independent assessments demonstrated the VEGP 3&4 Internal Flooding PRA substantially met the ASME/ANS PRA Standard RA-Sa-2009.

2.2 Resolution of F&Os from RG 1.200 Internal Flooding Peer Review

The VEGP 3&4 Internal Flooding PRA was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted in 2025. No findings remain open. The results of the independent F&O closure assessment are documented and available for audit.

3.0 Fire PRA Model

The fire PRA (FPRA) model is built to integrate with the internal events PRA model using fire PRA logic models (plant response model for CDF and LERF) and database inputs comprising identified fire scenarios, frequencies, and target sets. CDF and LERF results include contribution from all credible fire sources and unscreened fire compartments within the fire PRA global plant analysis boundary. Multi-unit impacts for VEGP 3&4 Fire PRA are limited.

- The AP1000 plant is a single independent plant. Unit 3 shares a switchyard with VEGP 1&2. The Unit 4 switchyard is separate. The transformer areas for Unit 3 and 4 are separate but in proximity.
- The plant does not share systems or components with another plant. Unit 3 and Unit 4 are located in proximity; however, the units do not share any components or cables. The separation of the standard plant units is a fundamental design criterion of the AP1000 plant units.
- The other operating units at the site are separate from the AP1000 plant units aside from the switchyard.

Elements from the ASME standard were included in the scope of self-assessments and peer reviews: Plant Partitioning (PP), Equipment Selection (ES), Cable Selection and Location (CS), Qualitative Screening (QLS), FPRA Plant Response Model (PRM), Fire Scenario Selection and Analysis (FSS), Ignition Frequency (IGN), Circuit Failures (CF), Human Reliability Analysis (HRA), Fire Risk Quantification (FQ), and Uncertainty and Sensitivity Analysis (UNC). Reference to these elements leads to a consistent application of industry consensus methods and data sources in FPRA development and model reviews.

3.1 RG 1.200 Previous and Current Peer Reviews – Fire PRA Model

The At-Power Fire PRA was peer reviewed against Part 4 of ASME/ANS RA-Sa-2009 in 2020. Since the 2020 peer review, the VEGP 3&4 plant At-Power Fire PRA has been updated to address several peer review F&Os, to the extent possible for an early operational plant. Follow on work addressed further model improvement focused on as-built, as-operated plant information.

Following further F&O closure efforts, model maintenance, and self-assessment in 2025, the VEGP 3 & 4 Fire PRA Model was subject to a focused-scope industry peer review conducted by the PWROG which included a closure review to address previous findings. The peer review and subsequent independent assessments demonstrated the VEGP 3&4 Fire PRA substantially met the ASME/ANS PRA standard RA-Sa-2009.

3.2 Resolution of F&Os from RG 1.200 Fire PRA Peer Review

The VEGP 3&4 PRA Fire Model was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted in 2025 with findings remaining as either out of scope (not applicable) or not fully resolved to meet Category II or better. The finding that remains open is summarized in Table A3-3 along with the associated disposition relative to the 10 CFR 50.69 program. The results of the independent F&O closure assessment are documented and available for audit.

Table A3-3 Summary of Peer Review Findings – Fire PRA			
F&O No.	**F&O Details**	**Associated Supporting Requirement (SR)**	**Impact on Application**
06-001	Corrections are indicated for an event tree that performs calculations for the non-suppression probability to ensure consistency with data sources.	FSS-C7 – SR MET	There is no impact on the 50.69 application. The values affected are occupancy and maintenance factors, reliability values supporting fire suppression systems, and a typographical error. The associated FSS notebook attachment was updated to address the discrepancies identified.

4.0 Technical Adequacy – Other External Hazard Screening

The information provided in this section demonstrates that the VEGP 3&4 other external events screening meets the expectations for PRA scope and technical adequacy as presented in NRC RG 1.200, Revision 3 (ADAMS Accession No. ML20238B871**Error! Reference source not found.**). Conservative analysis has been demonstrated through progressive screening and application of qualitative and quantitative criteria.

The ASME/ANS PRA Standard ASME/ANS RA-Sa-2009 contains a total of 17 numbered supporting requirements for High Level Requirements for Other External Hazards: Requirements for Screening and Conservative Analysis. During development and self-assessment, two additional SRs supporting high winds (WHA-1 and WHA-3) were determined to meet the ASME standard with respect to tornado analysis and frequency of high straight winds. All hazards were screened via this analysis; two hazards, Internal Flooding and Seismic documented as "Not Screenable" are addressed separately from this analysis via a full scope at-power PRA.**Error! Reference source not found.** For the 10 CFR 50.69 application, the EPRI alternative seismic approach will be used, as described in the enclosure.

4.1 RG 1.200 Previous and Current Peer Reviews – Other External Hazard Screening

The At-Power External Events analysis was peer reviewed against Part 6 of ASME/ANS RA-Sa-2009 in 2016. Since the 2016 peer review, the VEGP 3&4 At-Power external events PRA has been updated to address peer review F&Os. Seismic hazards and other significant external events with potential impact on risk are evaluated in accordance with the PRA standard to support the application.

The VEGP 3&4 Other External Hazard Screening was subject to a focused-scope industry peer review conducted by the PWROG following self-assessment in 2025. The peer review and subsequent independent assessments demonstrated the VEGP 3&4 Other External Hazards Screening met all Supporting Requirements at Capability Category (CC) II or better.

4.2 Resolution of F&Os from RG 1.200 Other External Hazard Screening Peer Review

The VEGP 3&4 PRA Other External Hazard Screening was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted in 2025, with no open findings remaining. The results of the independent F&O closure assessment are documented and available for audit.

5.0 General Conclusions Regarding PRA Capability

The information provided in this attachment demonstrates that the VEGP 3&4 PRA models conform to the Standard at CC II, which satisfies the guidance of RG 1.200, Revision 3.

The VEGP 3&4 internal events, internal flooding, and fire PRA models, and the Other External Hazard Screening technical capability evaluations described above provide a robust basis for concluding that the PRA models are suitable for use in risk-informed processes such as that proposed for 10 CFR 50.69 categorization.

Attachment 4: External Hazards Screening

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
Aircraft Impact	Y	C-C	An assessment performed concluded that an aircraft impact would not inhibit VEGP 3&4 core cooling capability, would not impact containment integrity and would not impact spent fuel pool integrity based on best-estimate calculations. Specifically, the assessment found that safety-related components inside containment, including the reactor pressure vessel and passive core cooling system, remain intact and maintain their intended capabilities following the shock-induced vibrations resulting from the impact of a large commercial aircraft.
Avalanche	Y	C-3	Topography, geography, and climate are such that no avalanche is possible.
Biological Events	Y	C-1 C-5	The Raw Water System (RWS) design features would prevent any impact on plant operation due to biological events. Additionally, there are pre-emptive as well as mitigative actions (i.e., usage of alarm response, sampling, and system operating procedures) that will prevent biological hazards from negatively impacting the circulating water system or service water system operation.
Coastal Erosion	Y	C-1 C-3	The Vogtle site is inland and not subject to coastal erosion. The general phenomenon of erosion was also considered as it applies to the banks of the Savannah River. The scenario of a bluff sliding into the river upstream or downstream, leading to redirection of flow or causing flooding around the intake structure was analyzed. As the safety-related cooling water systems for VEGP 3&4 do not require water from river intake, river intake would not be classified as safety-related structure and loss of river intake would have no adverse effects.
Drought	Y	C-1 C-5	VEGP 3&4 does not require a conventional ultimate heat sink to provide safety–related cooling during emergency shutdown. Consequently, river water will not be necessary to achieve safe shutdown of the units.

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
External Flooding	Y	C-1, C-A (individual hazard) C-B (combined hazard)	Various flooding scenarios were analyzed and the local Probable Maximum Precipitation (PMP)-based Probable Maximum Flood (PMF) analysis shows the detailed grading design does not cause flooding of any safety-related facilities at VEGP 3&4 and supports the licensing basis PMP-based PMF analysis. The potential cliff edge effect of a PMP event was also evaluated, and several factors prevent a cliff edge effect from presenting a risk to the site during a PMP event. Even if rainwater levels were to somehow exceed values determined in the PMP analysis, no water will enter containment or the auxiliary building because these structures have no outside doors at ground level.
Electromagnetic Interference	Y	C-1	AP1000 Safety-Related Systems subject to equipment qualification are designed to withstand electromagnetic interference conditions that would exist before, during and following a design basis accident without loss of safety function for the time required to perform the safety function. Examples include equipment in the PMS and DAS among others. Therefore, electromagnetic interference can be screened based on C-1.
Extreme Wind or Tornado	Y	C-C	The overall extreme winds and tornadoes hazard risk is calculated as the sum of the CDF calculated for straight winds and tornadoes and was calculated to be less than 1.00E-08 per year, thus this hazard screens out based on Criterion C-C.
Fog	Y	C-1 C-4	Heavy fog would not impact plant operations or the ability to achieve safe shutdown if necessary. Fog does affect the frequency of occurrence of other hazards (i.e. transportation accidents and aircraft operations) so it is indirectly considered.
Forest or Range Fire	Y	C-1 C-3	The surrounding plant terrain is characterized by gently rolling hills and is approximately 30% farmland, with the remainder primarily wooded areas. The toxic chemicals emitted from a forest fire are CO, NO2, and CH4. Calculations performed for Units 1 & 2 demonstrate that the pollutant concentrations outside the control room air intake for a variety of wind speeds (from 0.25 to 10 m/sec) and the Pasquill stability category G are effectively zero. Since Units 3 and 4 are approximately adjacent to Units 1 and 2 and the vegetation in the vicinity remains the same even after revegetation of the Units 3 and 4

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
			construction site, release of toxic combustion products from the onsite forest fire does not pose a hazard to the Units 3 and 4 control room operators. Calculations were also performed for heat flux and resultant temperature rise, temperature rise (~46.5°C) is less than the allowable rise (bulk 194°C and local 361°C). Therefore, a forest fire will not cause thermal damage to VEGP 3&4 safety-related structures, based on the distance from the forest (1836 ft away).
Frost	Y	C-1 C-4	Frost has lesser damage potential than snow and ice; therefore, the design parameters for snow and ice are bounding relative to those for frost.
Hail	Y	C-1 C-4	The consequences from hail are less severe than the tornado missiles for which the plant is designed.
High Summer Temperature	Y	C-1 C-5	VEGP 3&4 has a max safety temperature of 115° F dry bulb / 77.7° F wet bulb coincident and 83.9 °F wet bulb noncoincident, and max normal temperature of 97° F dry bulb / 76° F wet bulb coincident and 79°F wet bulb noncoincident which falls within the AP1000 plant DCD site parameters.
High Tide	Y	C-3	The VEGP 3&4 site is inland and not subject to this hazard.
Hurricane	Y	C-4	The VEGP 3&4 site is inland. Hurricane rainfall is included under external flooding and the hurricane wind effects are covered under extreme winds and tornados.
Ice Cover	Y	C-1	The Vogtle site specific characteristic for snow/ice loads is 10.0 pounds per square foot which falls within the AP1000 DCD site parameters which is 75 pounds per square foot.It is very unlikely that surface or frazil ice formation would occur in the Savannah River in the vicinity of the VEGP 3&4 river intake location. Even in the event of a very unlikely ice event on the Savannah River, there would be no impact on the ability to complete a safe plant shutdown if necessary.
Industrial or Military Facility Accident	Y	C-1 C-A	The locations of importance identified were the Savannah River site (SRS), Unitech Services Nuclear Laundry Facility, Chem-Nuclear Systems, Plant Wilson, VEGP 1&2, and Fort Gordon, which have been evaluated.

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
			There are no major depots within 5 miles of the Vogtle site and the only chemical storage areas within 5 miles are at the SRS and the Wilson combustion turbine plant. The original analysis (performed for Units 1 and 2) had determined that SRS had the potential to use chlorine and ammonia at the D-Area, but it was confirmed that this is no longer the case (only chlorine softener chemicals and biocide are used). There were no chemicals identified that would be hazardous to the Vogtle site or would require further evaluation. The chemicals stored at the Plant Wilson combustion turbine plant consist of fuel oil, sulfuric acid, and several other chemicals kept in small quantities. These chemicals have low volatility and toxicity, and there would be no potential hazard. The hazard due to the release of radioactive material from either VEGP 1&2 or the facilities at SRS, because of normal operations or an unanticipated event, would not threaten safety of the new units. Safety-related structures, systems, and components for the plant have been designed to withstand the effects of radiological events and the consequential releases which would bound the contamination from a release from either of these potential sources.
Internal Flooding	N	None	The internal flooding PRA model addresses risks from internal flood events.
Internal Fire	N	None	The internal fire PRA model addresses risks from internal fires.
Landslide	Y	C-3	Topography, geography, and climate are such that a landslide is not possible.
Lightning	Y	C-1 C-4	VEGP 3&4 includes a lightning protection system that provides protection of exposed structures and building housing safety-related and fire protection equipment. A lightning protection risk assessment was performed for the buildings comprising VEGP 3&4, and it was determined that the tolerable lightning frequency for each of the buildings was less than the expected lightning frequency; therefore, additional lightning protection was required as part of the design. The zone of protection is based on the elevations and geometry of the structures. It includes the space covered by a rolling sphere having a radius sufficient enough to cover the building to be protected. The zone of protection method is

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
			based on the use of ground masts, air terminals and shield wires. Either copper or aluminum is used for lightning protection. Lightning protection grounding is interconnected with the station or switchyard grounding system. Lightning strikes that result in a LOOP are accounted for in the generic LOOP IEF.
Low Lake Level or River Stage	Y	C-1 C-5	VEGP 3&4 does not require a conventional ultimate heat sink to provide safety–related cooling during emergency shutdown. Consequently, river water will not be necessary to achieve safe shutdown of the units. The only use of water from the Savannah River for the reactor units will be for the circulating water system/turbine plant cooling water system makeup, where river water will be required to replace evaporative water losses, drift losses, and blowdown discharge.
Low Winter Temperature	Y	C-1 C-5	The VEGP 3&4 minimum safety temperature is -8°F, and the minimum normal temperature is 21°F dry bulb which fall within the AP1000 DCD site parameters (minimum safety temperature of –40°F, and minimum normal temperature of -10°F). Additionally, there are proceduralized actions that mitigate the effects of low temperature, and these actions are undertaken any time outside temperatures drop below 35°F.
Meteorite or Satellite Impact	Y	C-C	The total risk from meteorite or satellite strike was calculated to be less than 1.00E-08/yr.
Pipeline Accident	Y	C-3	Three pipelines are within 25 miles of the VEGP 3&4 site, but none are located within 10 miles. The pipelines identified are well over 10 miles from the Vogtle site, it can be concluded that the hazard from pipeline accidents in terms of blast pressure and release of toxic chemicals is negligible since they cannot occur close enough to the plant to affect it.
Precipitation, Intense	Y	C-4	The impact of intense precipitation is included in external flooding.
Release of Chemicals in Onsite Storage	Y	C-1 C-A	Many of the chemicals stored on site are excluded from further consideration due to their properties (e.g., low volatility or low toxicity) or due to the relatively small quantities stored. An analysis concluded that for all releases except hydrazine, the average concentration over an 8-hour period would never exceed the long-term toxicity limit. The impact on Units 3 and 4 (located

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
			1800 ft from the hydrazine) due to an accidental hydrazine release is expected to be smaller than that for Units 1 and 2. Phosphoric acid (Nalco 3DT177) is one of the new chemicals used for the existing Units 1 and 2 that was identified to be toxic. Under stable atmospheric conditions the phosphoric acid concentration outside the new control room air intake would be 94 µg/m3, which is much lower than the 8-hour toxic threshold limit value of 1 mg/m3 and the short-term exposure limit of 3 mg/m3 following an accidental release. Since this material is not flammable, the explosion effect was not evaluated. Methoxypropylamine (MPA) was also evaluated for Units 1 and 2. The evaluation showed a concentration outside the control room of 1.5 ppm, compared to the allowable 15 ppm. Due to the distance between the new units and the existing units, the effects of accidental MPA release at Units 1 and 2 will be expected to be less at Units 3 and 4, and thus acceptable. The three 2000-gallon capacity above-ground liquid propane tanks are installed at the fire training facility on-site. The fire training facility propane tanks are located beyond the distance where an exploding tank or an exploding flammable vapor cloud would yield an overpressure of 1 psi on any safety-related structure. The fire training facility propane tanks were also evaluated for thermal effects from a fire. The calculated temperature rise is less than the maximum allowable temperature rise. The sources of onsite chemicals do not represent a toxic hazard to VEGP plant control room personnel.
River Diversion	Y	C-1 C-5	A potential diversion of the Savannah River from the river intake is unlikely to occur naturally. Also, analysis for VEGP 1&2 indicates that any possible effect on water supply to the intake from river channel diversion should come from extremely slow changes, which can be responded to as they occur. Additionally, the scenario of a bluff sliding into the river upstream or downstream, leading to redirection of flow or causing flooding around the intake structure was also analyzed. As the safety-related cooling water systems for VEGP 3&4 do not require water from river intake, river intake would not be classified as safety-related structure

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
			and loss of river intake would have no adverse effects.
Sandstorm	Y	C-3	Sandstorms are not applicable at the Vogtle site where the surrounding plant terrain is gently rolling hills consisting of approximately 30% farmland, with the remainder primarily wooded areas.
Seiche	Y	C-3	The Vogtle site is not located on a large, enclosed body of water therefore flooding due to seiche is precluded.
Seismic Activity	N	None	VEGP 3&4 is a Tier 1 site as defined by EPRI 3002017583. See Section 3.2.3 of the enclosure.
Snow	Y	C-C	The sensitivity study performed for all snow results in a value below the 1E-08/yr threshold.
Soil Shrink-Swell Consolidation	Y	C-1 C-3	The Upper Sand Stratum below the ground water for the Vogtle site is susceptible to soil liquefaction. To eliminate that concern, this sand layer was removed and replaced with engineered fill compacted to a minimum of 95% of the maximum dry density. Usage and compaction of the engineered fill eliminate susceptibility to shrink/swell, sinkhole, and land rise phenomena.
Solar Flares or Solar Storms	Y	C-C	Considering conservatism, the hazard can be screened with respect to 1E-06 per year.
Storm Surge	Y	C-4	The impact of storm surges is included in external flooding.
Toxic Gas	Y	C-4	The impact from release of toxic gas is included in industrial or military accidents, release of chemicals from on-site storage, and transportation accidents.
Transportation Accident	Y	C-3 C-A	Explosions and Flammable Vapor Clouds due to Truck Traffic: The closest distance from the site delivery route to the power block circle is approximately 2000 ft. For an explosion from a 4000-gallon truck, the 1 psi blast pressure distances and the vapor cloud explosion are 1340 ft and 1658 ft from the road, respectively, and so will have no impact on Unit 3 or 4. Since transported diesel/fuel oil is not flammable and is much less volatile than gasoline, the gasoline truck analysis becomes bounding in the evaluation of truck-borne hazards. Explosions and Flammable Vapor Clouds due to Railroad Traffic: For the explosion from a rail car, the TNT equivalent safe distance beyond which

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
			the blast pressure would be less than 1 psi was calculated to be 220 ft. For an explosion from a flammable vapor cloud, the TNT equivalent safe distance beyond which the blast pressure would be less than 1 psi was calculated to be 451 ft. The distance between the closest point of the rail line and Units 3 and 4 is approximately 4.5 miles which is far greater than either of the above calculated critical distances. Therefore, there will not be any impact on Unit 3 or 4 from an explosion of cyclohexane from a rail car or vapor cloud. Release of Hazardous Chemicals Due to a Transportation Accident: The evaluation for release of ammonia transported by rail car was originally performed for Units 1 and 2, and it can be extended to Units 3 and 4. Assuming the release from a rail car containing 26 tons of anhydrous ammonia, the evaluation showed that the Units 1 and 2 control room concentration at 2 minutes after odor detection is 112 ppm, without taking credit for control room isolation. This concentration is much lower than the IDLH value of 300 ppm. The evaluation assumed 2 minutes is sufficient time for a trained operator to put a self-contained breathing apparatus into operation, if they are to be used. In addition, the AP1000 plant design provides manual actuation to initiate the emergency habitability system.
Tsunami	Y	C-3	The Vogtle site is inland and not subject to this hazard.
Turbine-Generated Missiles	Y	C-B C-C	The following turbine-generators were evaluated: VEGP 3&4: An analysis was performed which concluded that the probability of destructive overspeed condition and missile generation, assuming the recommended inspection and test frequencies, is less than 1.00E-08 per year. In addition, orientation of the turbine-generator is such that a high-energy missile would be directed at a 90-degree angle away from safety-related structures, systems, or components. Plant Wilson: The combustion turbine plant is normally in standby and is only for peak load. There is a small hill with a wooded area between Plant Wilson and the VEGP 3&4 site which would prevent the turbine missile from reaching the site with any penetrable velocity. Additionally, the six (6) combustion turbines at Plant Wilson have a

External Hazard	Screening Result		
	Screened? (Y/N)	Screening Criterion (Note a)	Comment
			favorable orientation away from the VEGP 3&4 site. VEGP 1&2: The total risk to VEGP 3&4 from a missile generated by the Unit 1 or 2 turbine generators is less than the 1E-08/yr threshold.
Volcanic Activity	Y	C-3	The Vogtle site is not located near any active volcanoes.
Waves	Y	C-4	The impact of waves is included in external flooding.
Note a – See Attachment 5 for descriptions of the screening criteria.			

Attachment 5: Progressive Screening Approach for Addressing External Hazards

Screening Step	Criterion	Source/Comments
Initial Screening: **EXT-B1**	C-1 – Event damage potential is equal or lesser than that for which the plant is designed.	ASME/ANS Standard RA-Sa-2009
	C-2 – Event has lower mean frequency and no worse consequences than other events analyzed.	
	C-3 – Event cannot occur close enough to the plant to affect it.	
	C4 – Event is included in the definition of another event.	Note: Criterion C-4 is not used to screen. It is used only to include with another event.
	C-5 – Event develops slowly, allowing adequate time to eliminate or mitigate the threat.	
Progressive Screening: **EXT-C1**	C-A – Design basis hazard cannot lead to a core damage accident.	ASME/ANS Standard RA-Sa-2009
	C-B – Design-basis-hazard event has a mean frequency of < 10-7/yr, and the mean conditional core damage probability (CCDP) is < 0.1.	Note: The ASME C-B and C-C values were adjusted to consider the passive design of VEGP 3&4 which leads to CDF two orders of magnitude lower.
	C-C – Bounding mean CDF is < 10-8/yr.	

Attachment 6: Disposition of Key Assumptions/Sources of Uncertainty

1.0 Introduction

The Vogtle Electric Generating Plant, Units 3 and 4 (VEPG 3&4) Probabilistic Risk Assessment (PRA) modeling assumptions and sources of uncertainty were reviewed to identify those which could be significant for the 10 CFR 50.69 Program. The baseline internal events, internal flooding, and internal fire PRA models document assumptions and sources of uncertainty and these were reviewed during the respective hazard peer reviews. The approach taken is, therefore, to review these documents to identify the items which may be directly relevant to the 10 CFR 50.69 Program, evaluate and provide dispositions for the 10 CFR 50.69 Program, and perform sensitivity analyses if appropriate.

The list of assumptions and sources of uncertainty for all PRA models were reviewed to identify those which would be significant for the evaluation of this application. If the VEGP 3&4 PRA model used a non-conservative treatment, or methods that are not commonly accepted, the underlying assumption or source of uncertainty was reviewed to determine its impact on this application. To identify these assumptions and sources of uncertainty, both plant-specific and generic sources of uncertainty were considered. All PRA notebooks were reviewed, and sources of uncertainty were compiled and characterized in the uncertainties notebook for the internal events model and in the quantification notebooks for internal flooding and fire hazards. The identification and characterization of the sources of uncertainty was performed consistent with the requirements of the ASME/ANS PRA Standard (ASME/ANS RA-Sa-2009. This evaluation meets the intent of steps C-1 and E-1 of NUREG-1855, Revision 1. To assess the impact of sources of uncertainty, a review of the base case sources of uncertainty for the VEGP 3&4 PRA models was performed. Each identified uncertainty was evaluated with respect to its potential to significantly impact the decisions of this submittal. This evaluation meets the intent of the screening portion for steps C-2 and E-2 of NUREG-1855, Revision 1.

The characterization of assumptions and uncertainties for the baseline at-power PRA model is performed in associated notebooks. In general, the process used is as follows:

1. All plant-specific assumptions and uncertainties from the PRA notebooks are reviewed and dispositioned for potential impact on the baseline PRA model. At this stage, generic sources of uncertainty are included in the assessment.
2. Generic and plant-specific assumptions and sources of model uncertainty that have not been characterized in the individual PRA notebooks, are retained for further evaluation.
3. Any remaining assumptions and uncertainties (generic or plant-specific) are characterized, and applicable sensitivities are performed to assess impacts of those items on the baseline PRA model.

The characterization of the assumptions and uncertainties of the baseline models is generally representative of the impacts that those assumptions and uncertainties will have on the 10 CFR 50.69 Program. However, a complete review of these assumptions was performed to identify specific assumptions and sources of uncertainty that could have a unique impact on the 10 CFR 50.69 Program. This approach identifies applicable epistemic uncertainty impacts for

the internal events PRA and any epistemic uncertainty impacts that are unique to the internal flood and fire PRAs.

Key VEGP 3&4 PRA model specific assumptions and sources of uncertainty for this application were identified and dispositioned in the following sections. The conclusion of this review is that no additional sensitivity analyses are required to address the VEGP 3&4 PRA model specific assumptions or sources of uncertainty for the 10 CFR 50.69 Program.

2.0 Assessment of Uncertainty Impacts – Internal Events PRA

Table A6-2 documents a review of the VEGP 3&4 internal events sources of model uncertainty that were designated as having a potential impact on applications, and more specifically, as pertaining to 10 CFR 50.69.

3.0 Assessment of Uncertainty Impacts – Internal Flooding PRA

Table A6-3 documents a review of the VEGP 3&4 internal flooding sources of model uncertainty that were designated as having a potential impact on applications, and more specifically, as pertaining to 10 CFR 50.69.

4.0 Assessment of Uncertainty Impacts – Fire PRA

Table A6-4 documents a review of the VEGP 3&4 fire and translation (real time risk model) sources of model uncertainty that were designated as having a potential impact on applications, and more specifically, as pertaining to 10 CFR 50.69.

Table A6-2 Disposition of Key Sources of Uncertainty – Internal Events				
ID #	**Summary of Assumption / Uncertainty**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
17	Power recovery is not modeled for LOOP events.	Accident Sequence and Success Criteria	The treatment is conservative.	The treatment has a conservative impact on importance measures for safety related SSCs.
245	ADS Stages 2 and 3 may be required to open to achieve adequate RCS depressurization for successful operation of ADS Stage 4 and/or RCS injection. To simplify complexity of the model the more restrictive ADS Stages 2 and 3 success criteria (2 out of 4) were used for some sequences that only require 1 out of 4 valves.	Accident Sequence and Success Criteria	The treatment is conservative.	The treatment is conservative and should not impact categorization of SSCs via the 10 CFR 50.69 Program.
246	One ADS Stage 4 valve is required on the non-pressurizer loop to support passive injection. The ADS Stage 4 success criterion addresses the potential injection gap phenomena by requiring at least one open ADS Stage 4 valve on the non-pressurizer loop. This success criterion is applied for all ADS Stage 4 event tree tops and does not consider the status of ADS Stages 1-3. This negative impact on passive injection flow can occur when ADS Stage 4 and PXS injection are actuated following RCS depressurization by ADS Stages 1-3 from the top of the pressurizer. This phenomenon is a concern with only one or more Stage 1-3 valves open or for any RCS breaks in lines connected to the pressurizer.	Accident Sequence and Success Criteria	The treatment is conservative.	The treatment is conservative and should not impact categorization of SSCs via the 10 CFR 50.69 Program.

Table A6-2 Disposition of Key Sources of Uncertainty – Internal Events				
ID #	**Summary of Assumption / Uncertainty**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
115	When two or more ADS Stage 4 valves are required, it is assumed that at least one valve opens on the non-pressurizer loop. MAAP4 has a global RCS pressure model and lumped RCS nodalization which is not able to capture the injection gap phenomena. By operating at least one valve on the non-pressurizer loop, the phenomenon is avoided. Two ADS Stage 4 valves opening on the non-pressurizer loop will perform like one valve open on each loop.	Accident Sequence and Success Criteria	Realistic approach	The treatment is realistic and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
1	ADS Stage 1 is not modeled in the success criteria analyses, and it is assumed that, based on the limited depressurization for ADS Stage 1, the accident progression would not change if ADS Stage 1 was modeled.	Accident Sequence and Success Criteria	Realistic approach	The treatment is realistic and results in no PRA importance for ADS stage 1 valves. The 10 CFR 50.69 categorization process involves qualitative evaluation of all SSCs in the scope of the system being categorized.

Table A6-2 Disposition of Key Sources of Uncertainty – Internal Events				
ID #	**Summary of Assumption / Uncertainty**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
42	The mean values for the assumed instrumentation and control (I&C) CCF Beta factor variables have been increased by 10% to account for the increased uncertainty of the applicability of the generic CCF data, derived from mechanical equipment to digital I&C components. This could be overestimating the I&C CCF beta factors.	Component & System Failure Rates	Use of global CCF events is conservative for digital systems.	The treatments are conservative. These sources of uncertainty are adequately addressed by prescribed sensitivity studies discussed in NEI 00-04.
30	There is currently no accepted methodology for modeling and quantifying software reliability. However, three types of global software failure have been modeled: protection and safety monitoring system (PMS) CCF Software fail to actuate, PMS CCF Software spurious initiating event (IE) and plant control system (PLS) CCF Software fail to actuate.	Component & System Failure Rates	The assumption may result in overestimating the failure probability of I&C functions due to software failure.	
21	A simplified beta factor common cause model approach was used for instrumentation and control hardware failures and other areas where large groups are present. This approach will not underestimate the contribution of common cause failures to the results but may result in conservative results.	Component & System Failure Rates	Treatment is conservative.	

Table A6-2 Disposition of Key Sources of Uncertainty – Internal Events				
ID #	**Summary of Assumption / Uncertainty**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
182	The PMS model is based on representative signals. The signals modeled for reactor trip include high pressurizer pressure and overtemperature delta temperature (OTDT). Representative reactor trip signals are used based on the previous studies completed for the PWR Owners' Group (PWROG) and the previous PWROG studies are still valid for the AP1000 plant.	System Analysis	Realistic treatment	The treatment is realistic and should not impact the categorization of SSCs via the 10 CFR 50.69 Program. Reactor trip actuation signals will be provided by multiple sources (plant parameters) for all initiators. Given the multiple ways the reactor can be tripped, representative modeling is appropriate to address the functional level trip signal unavailability.
208	MCR-HVAC is representative of the dependency of loss of MCR HVAC on MCR operator actions associated with sensitive electronics and therefore is modeled with every human action that is credited from the MCR using PLS. MCR-HVAC is a transfer to the HVAC model and is representative of the loss of MCR HVAC and the PMS operator action to load shed.	System Analysis	Conservative treatment of human action when room cooling has failed.	The treatment is conservative and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
Section 7.3 of EPRI NP-6041	An assumed minimum joint human error probability (JHEP) value set to 1E-06 for human reliability analysis (HRA) dependency combination.	HRA	Treatment is realistic and consistent with industry consensus approaches.	The treatment is adequately addressed by prescribed sensitivity studies discussed in NEI 00-04. The use of a 1E-06 JHEP is consistent with industry guidance and accepted practice for internal events PRAs.

Table A6-3 Disposition of Key Sources of Uncertainty – Internal Flooding				
ID	**Summary of Assumption / Uncertainty**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
1	The frequency of major flood, flood, and spray events is based on pipe failure probabilities from the EPRI Internal Flooding pipe rupture frequencies outlined in EPRI Report 3002000079, Revision 3.	Internal Flooding / Frequencies	Realistic based on available information.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
2	All electrical equipment (i.e., switchgears, motor control centers, electrical cabinets, motor-driven pumps, fans, chillers, motor operated valves (MOVs), solenoid operated valves (SOVs), etc.) is assumed to be susceptible to a flood event and fails as a result of spray or submergence. An MOV is assumed to fail as-is, and an SOV is assumed to fail due to flood effects.	Internal Flooding / Component vulnerability	The assumption that electrical equipment is failed by spray or submergence, and that MOVs fail as-is, are consistent with EPRI 1019194. SOVs are assumed to fail in the de-energized position because the presence of flooding effects may interrupt the electrical input to the solenoid and thus would be expected to be de-energized.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
3	All flood-susceptible PRA related components within a flood area where the flood originates were assumed to fail as a result of spray, submergence, or the impact from a high-energy line break (i.e., humidity, condensation, temperature concerns, jet impingement, pipe whip, etc.). This approach is used as a starting point. Refinement based on specific component design (e.g., water-proofing) spatial effects, low pressure source potential or other reasonable judgment could be used for limiting the effect.	Internal Flooding / Impacted equipment	Conservative scenario approach is taken initially and the impacted equipment for a given propagation pathway can be refined for more realistic situations. This should not be a source of model uncertainty in most applications.	This conservative treatment should not adversely impact the categorization of SSCs via the 10 CFR 50.69 Program.
4	An assumed minimum JHEP value set to 1E-05 for HRA dependency combination.	Internal Flooding / HRA	Treatment is realistic and consistent with industry consensus approaches.	The assumption is adequately addressed by prescribed sensitivity studies discussed in NEI 00-04. The use of a 1E-05 JHEP is consistent with industry guidance

Table A6-3 Disposition of Key Sources of Uncertainty – Internal Flooding				
ID	**Summary of Assumption / Uncertainty**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
				and accepted practice for internal flooding PRAs. The impact on CDF and LERF from post-initiator operator actions and flood isolation actions is minimal.

Table A6-4 Disposition of Key Sources of Uncertainty – Fire				
ID	**Summary of Assumption / Uncertainty[1]**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
1	Plant Boundary Definition and Partitioning Standard Element: Partitioning can impact presentation and interpretation of results in context of single and multi-compartment scenario contributions.	Internal Fire / Analysis boundary and Partitioning	Consistent with NUREG/CR-6850	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program..
2	Fire PRA Component Selection Standard Element: The issue for component list accuracy is related to an inappropriate component list which can improperly attribute fire risk to the wrong areas.	Internal Fire / Equipment Selection	Consistent with NUREG/CR-6850 Components in the Fire PRA are either modeled or screened therefore no uncertainty associated with inclusion.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
3	Fire PRA Cable Selection Standard Element: The concerns for cable selection are: inaccuracies or inconsistencies between plant partitioning and cable locations, rules for identifying cables, and accuracy of cable and raceway routing information.	Internal Fire / Cable Selection	Consistent with NUREG/CR-6850 Vogtle 3 and 4 cable selection is based on the cable routing database. Pull tickets for cables are tracked for confirmation of cable routing.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
4	Qualitative Screening: Pre-mature screening concern for missing potentially important fire compartment analysis.	Internal Fire / Qualitative Screening	None. Qualitative screening was not implemented.	No impact on the categorization of SSCs via the 10 CFR 50.69 Program.
5	Fire-induced Risk Model: Quality issues include whether all relevant fire events and sequences were included in the model appropriately and whether the model contains all relevant equipment failures and human actions.	Internal Fire / Fire-induced Risk Model	Consistent with NUREG/CR-6850 Accuracy and completeness concerns are minimized by a formal independent technical review of the notebook.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.

Table A6-4 Disposition of Key Sources of Uncertainty – Fire				
ID	**Summary of Assumption / Uncertainty[1]**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
6	Fire Ignition Frequency: Mapping fire frequencies can be impacted by locations, equipment counts, weighting factors, and plant-specific fire experience.	Internal Fire / Fire Ignition Frequency	Consensus Method: The effect of the above assumption is to preserve total plant fire frequency for each ignition source type. This is part of the fundamental basis for development of the generic ignition frequencies as detailed in NUREG/CR-6850 Task 6.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
7	Quantitative Screening: Potential for inappropriate screening of fire compartments.	Internal Fire / Quantitative Screening	None. Quantitative screening was not implemented.	No impact on the categorization of SSCs via the 10 CFR 50.69 Program.
8	Scoping Fire Modeling: Premature or inappropriate screening out of potentially important fires or compartments.	Internal Fire / Scoping Fire Modeling	Conservative based on generic distributions and inclusion for mode detailed analysis in NUREG/CR-6850 Task 11 or other iterations.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
9	Detailed Circuit Analysis: Analysis may be subject to judgement or inaccuracies if not addressed properly using established rule set.	Internal Fire / Detailed Circuit Analysis	Consistent with NUREG/CR-6850 Accuracy and completeness concerns are minimized by a formal independent technical review of the notebook.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
10	Circuit Failure Mode Likelihood Analysis: Whether the test data is applicable to the actual situation, application of the rules, and actual analysis performed properly.	Internal Fire / Circuit Failure Mode Likelihood Analysis	Consistent with NUREG/CR-6850 (while utilizing data from NUREG/CR-7150) Uncertainties in spurious probabilities and hot short durations are propagated in the parametric uncertainty analysis using the NUREG/CR-7150 provided beta distributions.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.

Table A6-4 Disposition of Key Sources of Uncertainty – Fire				
ID	**Summary of Assumption / Uncertainty[1]**	**Part of Model Affected**	**Generic Impact on Risk Applications**	**Evaluation of 10 CFR 50.69 Impact**
11	Detailed Fire Modeling: Assigned target failure probabilities including: proper modeling objectives, fire scenarios sufficiently represent fire risk profiles, compartment's characteristics, HRR, target location, fire modeling tool selection.	Internal Fire / Detailed Fire Modeling	Consistent with NUREG/CR-6850 Accuracy and completeness concerns are minimized by a formal independent technical review of the notebook.	The treatment is a consensus approach and should not impact the categorization of SSCs via the 10 CFR 50.69 Program.
12	Post-Fire Human Reliability Analysis: Same technical issues as any HRA (modeling sufficiently complete and appropriate, performance shaping factors and fire-type scenarios).	Internal Fire / Post-Fire Human Reliability Analysis	Consistent with NUREG-1921. HEPs modeled in detail. Accuracy and completeness concerns are minimized by a formal independent technical review of the notebook. An assumed minimum joint HEP (JHEP) value set to 1E-05 for HRA dependency combination.	The treatment is adequately addressed by prescribed sensitivity studies discussed in NEI 00-04. The use of a 1E-05 JHEP is consistent with industry guidance and accepted practice for internal fire PRAs. The impact on overall fire PRA results is minimal.